
OPTICAL FORCES AND TORQUES ON ATOMS

It has been known for some time that the interaction of laser light with a free atom can give rise to electromagnetic pressure forces that act on its centre of mass. These forces have been extensively examined both theoretically and experimentally: see, for example Metcalf and van der Straten (5.1) or Adams and Riis (5.2). A simple model of a two-level atom subject to a plane electromagnetic wave may be shown to give rise to two kinds of force acting on the atom centre of mass. These are, a dissipative force arising from the absorption of the light by the atom and its subsequent spontaneous emission in a random direction and a dipole force arising from the non-uniformity of the field distribution which can attract the atom to the regions of intense field. These forces underpin the manipulation of atoms by lasers in a variety of beam configurations. The dissipative force has been exploited in the Doppler cooling of the atomic motion (5.3, 5.4) and the dipole force used for trapping (5.5), while a combination of these effects has led to the realisation of Bose–Einstein condensation (5.6). These forces are, of course, closely related to the ones discussed in Section 4.

It appeared probable that the interaction of atoms with beams possessing orbital angular momentum should lead to new effects. Consequently, theoretical studies of the interaction of light with atoms have been conducted to examine how the main features of Doppler cooling and trapping are modified when a plane wave or a fundamental Gaussian beam is replaced with Laguerre–Gaussian light.

Allen *et al.* (5.7, **Paper 5.1**), studied the effects of the orbital angular momentum of light on atoms by considering the theory of forces due to Laguerre–Gaussian light and their effects on a two-level atom. In order to explore more fully the effects of orbital angular momentum on atomic motion, they extended their investigations to more than one beam. The theory was developed in terms of the optical Bloch equations following Dalibard and Cohen-Tannoudji (5.8), which allows the *ab initio* inclusion of relaxation effects and incorporates saturation phenomena. The solution of the optical Bloch equations in the adiabatic, or constant velocity, approximation gave insight into the time evolution of angular momentum effects for an atom in a coherent light beam assumed to have a complex amplitude and a Laguerre–Gaussian spatial distribution. In this paper, use is made of the semi-classical approximation by replacing the position and momentum operators and by their expectation values, while maintaining a quantum treatment for the internal dynamics of the atom. The validity of the semi-classical approximation requires that the spatial extent of the atomic wave-packet be much smaller than the wavelength of the radiation field and that the uncertainty in the

Doppler shift be much smaller than the upper state linewidth of the atom. This is the case for most atoms if the recoil energy of the atom is much smaller than the upper state linewidth. Within the semi-classical approximation, the atomic density matrix can be written readily and its time evolution determined in accordance with the Heisenberg equation of motion. The paper specifically investigates the motion of a magnesium ion for a number of different beam configurations. Perhaps the most interesting aspect of the motion is that which arises from the reciprocal interplay between motions in orthogonal directions.

Related work in this area includes that of Babiker *et al.* (5.9) who show that a two-level atom moving in a Laguerre–Gaussian beam is subject to a light induced torque about the beam axis directly proportional to the orbital angular momentum. This torque reduces in the saturation limit to the simple form $T = \hbar \ell \Gamma$, where 2Γ is the decay rate of the excited state. A proposed experiment to observe the torque on the atom has been published by Power and Thompson (5.10). Allen *et al.* (5.11, **Paper 6.5**) showed that an atom moving in a Laguerre–Gaussian beam experiences an azimuthal shift in resonant frequency of $\ell V_\phi / r$ where r is the radial atomic position and V_ϕ the azimuthal component of velocity. Lembessis (5.12) has considered an atom moving in a linearly polarised beam and included the Roentgen term which arises from the motion of the electric dipole interacting with the magnetic part of the optical field. He found terms which couple the photon angular momentum with that of the atomic particle. The other interesting aspect of this type of investigation is that of Allen, Lembessis and Babiker (5.13), who recognised that the azimuthal component of the dissipative force contains a term which is proportional to $\sigma \ell$. Clearly, just as reversing the sign of the orbital angular momentum changes the direction of the force, so too does changing the handedness of the circular polarisation of the light. Normally the handedness of circularly polarised light would not be expected to determine the gross motion of the atom, only its internal state. Although the term is small and is comparable with other small terms ignored in trapping calculations, it is none the less an example of spin–orbit coupling in light. We are used to the spin–orbit coupling of electrons but not of light. Liu and Milburn (5.14) investigated the classical 2-D nonlinear dynamics of cold atoms in far-off-resonant Laguerre–Gaussian beams. They showed that chaotic developments exist provided $\ell > 1$, when the beam is periodically modulated. The atoms are predicted to accumulate on several ring regions when the system enters a regime of global chaos.

It has been recognised for a very long time that although dipole radiation dominates the physics of the interaction of light and matter, other higher order processes are also possible. This is particularly important as the selection rules of quadrupole and higher order radiation explicitly have both spin and orbital angular momentum contributions. Until the use of Laguerre–Gaussian beams the only higher order processes observed were ones where it was not possible, even in principle, to observe a change in the orbital angular momentum of the light. In a study of the selection rules and centre-of-mass motion of ultra-cold atoms, van Enk (5.15, **Paper 5.2**) considered how spin and orbital angular momentum might be transferred to internal and external angular momentum of an atom in dipole and quadrupole transitions.

For a cold atom the recoil effect on the atom of even a single photon cannot be neglected and the state of the atom may become changed. When the atom is sufficiently cold the external atomic motion can no longer be described classically. Indeed, when its de Broglie wavelength becomes comparable to the wavelength of the light, the external motion must also be quantised. This is the situation analysed by van Enk who shows

that the spin and orbital angular momenta are distributed over the internal and external angular momenta of the atom. The key feature of this work, as with other work on the torque and forces on atoms, is that the transition rate depends on the specific spatial dependence of the field, while the internal selection rules are independent of the mode structure and do not depend on the external atomic motion.

The most successful, and most interesting, experiment concerning the interaction of orbital angular momentum and atoms is a fascinating one on cold caesium atoms conducted by Tabosa and Petrov (5.16, **Paper 5.3**). They demonstrated, using a non-degenerate four-wave mixing process, that orbital angular momentum can be transferred from the optical beam to the cold caesium atoms. They point out that this process is similar to the phenomenon of optical pumping where orientation and alignment is induced in atomic systems by means of polarised light, or spin angular momentum. Their observation, therefore, is the first example of optical pumping of the orbital angular momentum of light. The remarkable aspect of their experiment is that they observed the interchange of orbital angular momentum between two waves of different frequency as a result of pumping the caesium atoms. They argued that the low temperature of laser cooled atoms, as well as diminishing the Doppler broadening of the radiation, could also give access to the direct observation of mechanical effects induced by an atom-field interaction. However, as yet, experiments to observe any net transfer of angular momentum to the atomic system during the lifetime of the process remain to be carried out. At the time of writing it would appear that, as the orbital angular momentum is conserved, radiation processes of higher order than dipole should demonstrate mode conversion when both orbital and spin selection rules become invoked.

The characteristic shape of the intensity distribution of Laguerre–Gaussian modes, irrespective of their orbital angular momentum, make them ideal for interactions with atoms and for the creation of traps with dark zones in them. It is not surprising therefore that this topic has a fast growing literature. It is not, however, an area of work relevant to the interests of this volume.

Just as the intensity distribution of Laguerre–Gaussian beams suggest annular trapping schemes, it follows that such beams can also be used for the creation of Bose–Einstein condensates; see for example (5.17, 5.18). The effect of what occurs when the orbital angular momentum aspects of the beams are exploited, is yet to be studied.

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Atom dynamics in multiple Laguerre-Gaussian beams

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(Received 10 April 1996)

The leading radiation forces acting on an atom or ion subject to linearly polarized Laguerre-Gaussian (LG) light are studied. Particular emphasis is laid on the orbital angular momentum effects associated with LG light. The optical Bloch equations appropriate for the adiabatic approximation are derived and used to evaluate the forces and associated torque governing the atomic motion. The steady-state dynamics of the atom are explored for atoms subject to a single beam and multiple independent counterpropagating beams. The main features responsible for the dynamics of the atom, together with the dipole potentials characteristic of Laguerre-Gaussian light, are identified and discussed. The theory is illustrated by the numerical integration of the equation of motion for Mg^+ ions in various beam configurations. This yields information on trajectories, velocity evolution, and vibrational frequencies at potential minima. Interesting effects involving a reciprocal interplay between motions in orthogonal directions are demonstrated. Such features are purely dependent on the orbital angular momentum property of the light. Their possible use in controlling atomic motion is investigated. [S1050-2947(96)07210-1]

PACS number(s): 32.80.Pj, 42.50.Vk

I. INTRODUCTION

The radiation forces associated with the near-resonant interaction of laser light with atoms and ions have been the subject of intensive theoretical and experimental study [1–4] since the basic mechanisms were first recognized [5]. The simplest features can be described with reference to a two-level atom subject to an electromagnetic wave. Near resonance, such an atom experiences two distinct forces: a dissipative force that arises from the absorption of the light by the atom followed by its spontaneous emission and a dipole force that arises from the nonuniformity of the field distribution. These basic forces underpin many of the applications involving the manipulation of atoms by lasers in a variety of beam configurations. The dissipative force has been exploited in cooling the atomic motion [6] and the dipole force used for trapping [7].

Much of the previous theoretical work in this context has assumed plane-wave modes. However, the demonstration that Laguerre-Gaussian (LG) laser beams possess well-defined orbital angular momentum $\ell\hbar$ [8,9] that originates in the azimuthal phase dependence of the field distribution has aroused new interest in the basic physics. The orbital angular momentum of LG beams is quite distinct from the spin angular momentum associated with circularly polarized light and can occur in linearly polarized LG modes. A circularly polarized LG beam possesses spin angular momentum as well as orbital angular momentum and can exhibit features involving spin-orbit coupling [10]. In our recent work [11] we presented a theory for the motion of a two-level atom in a Laguerre-Gaussian beam with spontaneous emission and saturation effects taken into account heuristically. The results found were in the form of an azimuthal shift in the atomic resonance and a torque about the beam axis.

The purpose of this paper is twofold: first, to present a more rigorous theory for the forces due to LG light and their effects on a two-level atom and, second, we extend our investigations on the orbital angular momentum effects beyond

the one beam case in order to explore more fully the effects of the orbital angular momentum on atomic motion. The theory is developed in terms of the optical Bloch equations (OBEs) [1,12,13] that allow the *ab initio* inclusion of relaxation effects and naturally incorporate saturation phenomena. The solution of the OBEs in the adiabatic, or constant velocity, approximation lends insight into the time evolution of angular momentum effects for an atom in a LG beam. In the long-time limit, these solutions lead directly to the steady-state results for the dissipative and dipole forces due to a LG beam.

In Sec. II we set up the density-matrix formalism appropriate for a two-level atom interacting with Laguerre-Gaussian light. In the adiabatic approximation this leads to the optical Bloch equations that formally enable the calculation of the average mechanical force (defined as the rate of change of the atomic momentum). We argue that as the concept of force is a classical one, the rate of change of momentum can only be interpreted as a force for elapsed times greater than the spontaneous decay time Γ^{-1} , where time is measured from the instant the light beam is switched on [1]. With this restriction, we solve the OBEs to determine the evolution of the average “force” components from the instant the LG beam is switched on. It is possible to examine the time evolution in a number of limits, but we discuss primarily the steady state. Further insight into the nature of the forces is gained by the numerical solution of the optical Bloch equations for a typical set of parameters. In Sec. III we give an analysis of the steady-state dissipative and dipole forces for various beam configurations and identify the features directly attributable to the angular momentum of the LG beams. The motion of a Mg^+ ion in counterpropagating LG fields is described in Sec. IV after solving the equation of motion numerically. The results demonstrate the effects of a characteristic torque and of reciprocating forces between axial and azimuthal motions. Section V contains conclusions and further comment.

II. FORMALISM

We wish to examine the evolution of the average force acting on a two-level atom or ion, henceforth referred to as the atom, due to its interaction with light. The light is in the form of a coherent beam with a complex amplitude α and has a LG distribution [8]. An appropriate Hamiltonian is given by

$$H = H_A + H_F + H_{\text{int}}, \quad (1)$$

where H_A and H_F are the unperturbed Hamiltonians for the atom and field, respectively, and are

$$H_A = \frac{\mathbf{P}^2}{2M} + \hbar \omega_0 \pi^\dagger \pi, \quad (2)$$

$$H_F = \hbar \omega a^\dagger a. \quad (3)$$

Here $\mathbf{P}$ is the center-of-mass momentum operator and π and $\pi^\dagger$ are the atomic lowering and raising operators; M is the mass of the atom and ω_0 is the atomic transition frequency. In Eq. (3) a and $a^\dagger$ are the annihilation and creation operators and ω is the frequency of the field.

The interaction Hamiltonian H_{int} in Eq. (1) describes the coupling of the atom to the electromagnetic field and is given in the electric dipole approximation by

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}(\mathbf{R}), \quad (4)$$

where $\mathbf{d}$ is the atomic dipole moment operator and $\mathbf{E}(\mathbf{R})$ is the electric field evaluated at the position $\mathbf{R}$ of the atom. The atomic dipole moment operator may be written as

$$\mathbf{d} = \mathbf{D}_{12}(\pi + \pi^\dagger), \quad (5)$$

where $\mathbf{D}_{12}$ is the dipole matrix element of the atomic transition. The electric-field vector associated with a Laguerre-Gaussian mode propagating along the z axis is given by

$$\mathbf{E}(\mathbf{R}) = i[a \hat{\mathbf{e}} \mathcal{E}_{klp}(\mathbf{R}) e^{i\Theta_{klp}(\mathbf{R})} - \text{H.c.}], \quad (6)$$

where $\hat{\mathbf{e}}$ is a polarization vector in the x - y plane. The electric field of a LG beam has a small vector component along the z axis [11,14], which we have ignored. It can easily be shown that the ignored term is of the order λ/w_0 relative to the principal component along $\hat{\mathbf{e}}$ [11]. In Eq. (6) $\mathcal{E}_{klp}(\mathbf{R})$ and $\Theta_{klp}(\mathbf{R})$ are, respectively, the mode amplitude and phase of the electric field, which may be written as [11,14]

$$\mathcal{E}_{klp}(\mathbf{R}) = \mathcal{E}_{k00} \frac{C_{lp}}{(1 + z^2/z_R^2)^{1/2}} \left(\frac{\sqrt{2}r}{w(z)} \right)^{|l|} L_p^{|l|} \left(\frac{2r^2}{w^2(z)} \right) e^{-r^2/w^2(z)}, \quad (7)$$

$$\Theta_{klp}(\mathbf{R}) = \frac{kr^2 z}{2(z^2 + z_R^2)} + l\phi + (2p + l + 1)\tan^{-1}(z/z_R) + kz. \quad (8)$$

Here $C_{lp} = \sqrt{p!/(|l|+p)!}$ is a normalization constant; $w(z)$ is given by $w^2(z) = 2(z^2 + z_R^2)/kz_R$, where z_R is the Rayleigh range. The integers l and p are indices characterizing the LG mode. It has been shown [8] that $\hbar h$ represents the orbital angular momentum of each quantum in the mode. Finally, in Eq. (7) $\mathcal{E}_{k00}$ corresponds to the plane-wave ampli-

tude for an axial wave vector k . The plane-wave amplitude and phase emerge directly from Eqs. (7) and (8) by setting $l=0$, $p=0$, and $z_R \rightarrow \infty$.

We now transform to an interaction picture with respect to the unperturbed field Hamiltonian $\hbar \omega a^\dagger a$. The field annihilation and creation operators then acquire the time dependences

$$a(t) = e^{i\omega a^\dagger a t} a e^{-i\omega a^\dagger a t} = a e^{-i\omega t}, \quad (9)$$

$$a^\dagger(t) = e^{i\omega a^\dagger a t} a^\dagger e^{-i\omega a^\dagger a t} = a^\dagger e^{i\omega t}. \quad (10)$$

In the classical limit in which the field forms a coherent beam, we may replace the field operators by c numbers

$$a(t) \rightarrow \alpha e^{-i\omega t}, \quad (11)$$

$$a^\dagger(t) \rightarrow \alpha^* e^{i\omega t}. \quad (12)$$

The coupling between the atom and field may then be written as

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}(\mathbf{R}) = -i\hbar[\tilde{\pi}^\dagger \alpha f(\mathbf{R}) - \text{H.c.}], \quad (13)$$

where in writing Eq. (13) we have made use of the rotating-wave approximation and have defined

$$\tilde{\pi} = \pi e^{i\omega t}, \quad (14)$$

$$f(\mathbf{R}) = (\mathbf{D}_{12} \cdot \hat{\mathbf{e}}) \mathcal{E}(\mathbf{R}) e^{i\Theta(\mathbf{R})/\hbar}. \quad (15)$$

For convenience, we have not explicitly shown the LG labels klp associated with $\mathcal{E}_{klp}(\mathbf{R})$ and $\Theta_{klp}(\mathbf{R})$ by virtue of Eqs. (7) and (8). In the rest of this section we continue to use this simple notation but resort to the full notation subsequently.

To derive the optical Bloch equations for the atomic density-matrix elements we make the assumption that the position and momentum operators $\mathbf{R}$ and $\mathbf{P}$ may be replaced by their expectation values $\mathbf{R}_0$ and $\mathbf{P}_0$, respectively. This approximation allows the gross motion of the atom to be treated classically, while maintaining a quantum treatment for the internal dynamics of the atom. The validity of the semiclassical approximation requires that the spatial extent of the atomic wave packet be much smaller than the wavelength of the radiation field and that the uncertainty in the Doppler shift be much smaller than the upper-state linewidth of the atom. This is the case for most atoms [1] if the recoil energy of the atom is much smaller than the upper-state linewidth.

Within the semiclassical approximation, the atomic density matrix can be written as

$$\rho = \delta(\mathbf{R} - \mathbf{R}_0) \delta(\mathbf{P} - \mathbf{P}_0) \rho(t), \quad (16)$$

where the internal dynamics of the atom are now contained in $\rho(t)$. The evolution of $\rho(t)$ is given by the well-known relation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] + \mathcal{R}\rho, \quad (17)$$

where $\mathcal{R}\rho$ accounts for the relaxation dynamics of the atomic system. By substitution of H and use of the coupling given in

Eq. (13), we obtain the following optical Bloch equations for the atomic density-matrix elements:

$$\frac{d\rho_{22}}{dt} = -2\Gamma\rho_{22} - \alpha f(\mathbf{R}_0)\tilde{\rho}_{12} - \alpha^* f^*(\mathbf{R}_0)\tilde{\rho}_{21}, \quad (18)$$

$$\frac{d\tilde{\rho}_{21}}{dt} = -(\Gamma - i\Delta_0)\tilde{\rho}_{21} + \alpha f(\mathbf{R}_0)(\rho_{22} - \rho_{11}), \quad (19)$$

where $\Delta_0 = \omega - \omega_0$ is the detuning of the field frequency from atomic resonance and $\tilde{\rho}_{21} = \langle \tilde{\pi} \rangle$.

The average radiation force acting on the atom is defined as the average rate of change of the atomic momentum. We may write

$$\langle \mathbf{F} \rangle = -\langle \nabla H_{\text{int}} \rangle. \quad (20)$$

Substitution of Eq. (13) into Eq. (20) and use of Eq. (15) allows the force to be written as $\langle \mathbf{F} \rangle = \langle \mathbf{F}_{\text{diss}} \rangle + \langle \mathbf{F}_{\text{dipole}} \rangle$. Here $\langle \mathbf{F}_{\text{diss}} \rangle$ is the dissipative force given by

$$\langle \mathbf{F}_{\text{diss}} \rangle = -\hbar \nabla \Theta(\mathbf{R}_0) \{ \tilde{\rho}_{12}(t) \alpha f(\mathbf{R}_0) + \tilde{\rho}_{21}(t) \alpha^* f^*(\mathbf{R}_0) \} \quad (21)$$

and $\langle \mathbf{F}_{\text{dipole}} \rangle$ is the dipole force given by

$$\langle \mathbf{F}_{\text{dipole}} \rangle = i\hbar \frac{\nabla \Omega(\mathbf{R}_0)}{\Omega(\mathbf{R}_0)} \{ \tilde{\rho}_{12}(t) \alpha f(\mathbf{R}_0) - \tilde{\rho}_{21}(t) \alpha^* f^*(\mathbf{R}_0) \}, \quad (22)$$

where we have introduced a position-dependent Rabi frequency as $\Omega(\mathbf{R}_0) = |\alpha(\mathbf{D}_{12} \cdot \hat{\mathbf{e}}) \mathcal{E}(\mathbf{R}_0)|/\hbar$.

In the adiabatic approximation [15], the atomic velocity $\mathbf{V} = \mathbf{p}_0/M$ is assumed to be constant during the time taken for the dipole moment to relax to its steady-state value. The position $\mathbf{R}_0$ of the atom at time t is then given by

$$\mathbf{R}_0 = \mathbf{r}_0 + \mathbf{V}t, \quad (23)$$

where $\mathbf{r}_0$ is the (initial) position of the atom when the beam was switched on. Thus we can write

$$f(\mathbf{R}_0) = f(\mathbf{r}_0 + \mathbf{V}t) \quad (24)$$

$$\approx f(\mathbf{r}_0) e^{i \nabla \Theta(\mathbf{r}_0) \cdot \mathbf{V}t}, \quad (25)$$

where we have assumed that the change in the field amplitude is negligible during the time taken for the dipole moment to relax to its steady-state value.

Within the adiabatic approximation, the optical Bloch equations take the form

$$\frac{d\rho_{22}}{dt} = -2\Gamma\rho_{22} - \alpha f(\mathbf{r}_0)\hat{\rho}_{12} - \alpha^* f^*(\mathbf{r}_0)\rho_{22}, \quad (26)$$

$$\frac{d\hat{\rho}_{21}}{dt} = -[\Gamma - i\Delta(\mathbf{r}_0, \mathbf{V})]\hat{\rho}_{21} + \alpha f(\mathbf{r}_0)(\rho_{22} - \rho_{11}), \quad (27)$$

where the total detuning $\Delta(\mathbf{r}_0, \mathbf{V}) = \Delta_0 - \nabla \Theta(\mathbf{r}_0) \cdot \mathbf{V}$ and $\hat{\rho}_{21} = \tilde{\rho}_{21} e^{-i \mathbf{V} \cdot \nabla \Theta(\mathbf{r}_0)}$. The forces can now be written as

$$\langle \mathbf{F}_{\text{diss}} \rangle = -\hbar \nabla \Theta(\mathbf{r}_0) \{ \hat{\rho}_{12}(t) \alpha f(\mathbf{r}_0) + \hat{\rho}_{21} \alpha^* f^*(\mathbf{r}_0) \}, \quad (28)$$

$$\langle \mathbf{F}_{\text{dipole}} \rangle = i\hbar \frac{\nabla \Omega(\mathbf{r}_0)}{\Omega(\mathbf{r}_0)} \{ \hat{\rho}_{12}(t) \alpha f(\mathbf{r}_0) - \hat{\rho}_{21}(t) \alpha^* f^*(\mathbf{r}_0) \}. \quad (29)$$

For given initial conditions the solution of the optical Bloch equations (26) and (27) leads formally to the determination of the forces by direct substitution in Eqs. (28) and (29).

A. Steady state

The steady state occurs when all time derivatives in the optical Bloch equations are set equal to zero and corresponds to the long-time limit. It is not difficult to show that the steady-state solutions to the optical Bloch equations (26) and (27) yield the following expressions for the steady-state forces:

$$\langle \mathbf{F} \rangle = \langle \mathbf{F}_{\text{diss}} \rangle + \langle \mathbf{F}_{\text{dipole}} \rangle, \quad (30)$$

where

$$\langle \mathbf{F}_{\text{diss}}(\mathbf{R}, \mathbf{V}) \rangle = 2\hbar \Gamma \Omega^2(\mathbf{R}) \left(\frac{\nabla \Theta(\mathbf{R})}{\Delta^2(\mathbf{R}, \mathbf{V}) + 2\Omega^2(\mathbf{R}) + \Gamma^2} \right), \quad (31)$$

$$\langle \mathbf{F}_{\text{dipole}}(\mathbf{R}, \mathbf{V}) \rangle = -2\hbar \Omega(\mathbf{R}) \nabla \Omega(\mathbf{R}) \times \left(\frac{\Delta(\mathbf{R}, \mathbf{V})}{\Delta^2(\mathbf{R}, \mathbf{V}) + 2\Omega^2(\mathbf{R}) + \Gamma^2} \right), \quad (32)$$

where we have redefined the notation such that $\mathbf{R}$ now stands for the position of the atom (instead of $\mathbf{r}_0$). The above results are the same as those presented in our previous work where perturbation techniques for time-dependent Heisenberg operators have been used [11]. The dependence on the decay constant and on saturation are in agreement with our earlier heuristic approach.

B. Transients

Torrey [16] gave detailed solutions of the original optical Bloch equations. He also recognized that there were three special cases of interest that have relatively simple solutions. These were for strong collisions when the natural lifetime of the state may be replaced by the collision shortened lifetime, exact resonance and for intense external fields. His approach was applied by Allen and Eberly [17] to the optical Bloch equations. Consequently, in a similar way, the evolution of the forces from the instant the light beam is switched on can also be examined for a number of special cases. In fact, radiation effects have been examined in detail for atoms excited by plane-wave light [18]; the cases considered were (i) an atom with all relaxation constants equal to zero, (ii) a weak beam, (iii) exact resonance, and (iv) steady state achieved by an intense field. This treatment may be readily generalized for Laguerre-Gaussian light.

We shall settle simply for the steady-state case already considered in Sec. II A because the general time dependence of the density-matrix elements can be determined more readily for arbitrary parameter values by the numerical solution of the optical Bloch equations (26) and (27). This enables the evolution of the corresponding forces to be displayed. We display the results for a Laguerre-Gaussian mode

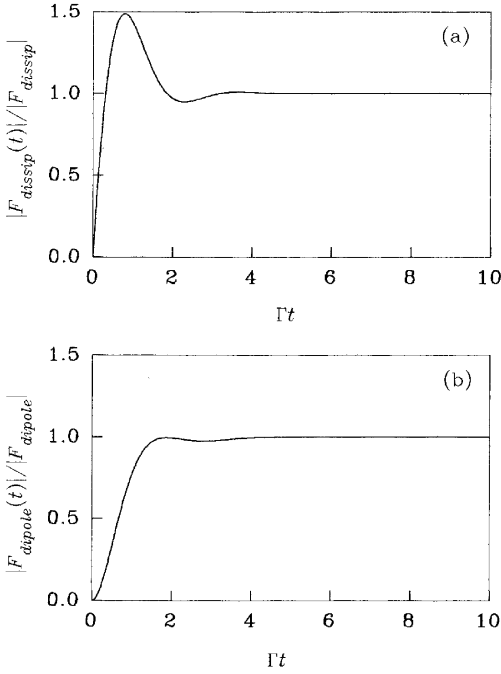


FIG. 1. Variation with time (in units of Γ^{-1}) of (a) the average dissipative force and (b) the average dipole force in units of the corresponding steady-state values for a stationary atom in a single LG beam. [The time variation of the corresponding torque would be the same as in (a), but in units of the steady-state torque.] See the text for the values assumed for the parameters.

with $l=1$ and $p=0$ such that $\Omega(\mathbf{r}_0)=\Gamma$; $\Delta=-\Gamma$ and $\omega(0)=35\lambda$, where $\lambda=280$ nm is the atomic transition wavelength. The results shown in Figs. 1(a) and 1(b) depict the evolutions of the dissipative and dipole forces [given by Eqs. (28) and (29), respectively]. These figures show clearly that the dipole moment, and hence force components, relax to their steady-state values within a time of the order of Γ^{-1} .

III. STEADY-STATE DYNAMICS

A. Single LG beam

From the results of the preceding section it is clear that, for elapsed times greater than the inverse relaxation parameters, the total force on a two-level atom has a steady-state value, exhibits position dependence, and is naturally divisible into two terms. Restoring the explicit reference to a specific Laguerre-Gaussian mode, the steady-state force on a moving atom due to a single Laguerre-Gaussian beam propagating along the positive z axis is written

$$\langle \mathbf{F} \rangle_{klp} = \langle \mathbf{F}_{\text{diss}} \rangle_{klp} + \langle \mathbf{F}_{\text{dipole}} \rangle_{klp}, \quad (33)$$

where

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}(\mathbf{R}, \mathbf{V}) \rangle_{klp} \\ = 2\hbar\Gamma\Omega_{klp}^2(\mathbf{R}) \left(\frac{\nabla\Theta_{klp}(\mathbf{R})}{\Delta_{klp}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \right) \end{aligned} \quad (34)$$

and

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}(\mathbf{R}, \mathbf{V}) \rangle_{klp} \\ = -2\hbar\Omega_{klp}(\mathbf{R})\nabla\Omega_{klp} \left(\frac{\Delta_{klp}(\mathbf{R}, \mathbf{V})}{\Delta_{klp}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \right), \end{aligned} \quad (35)$$

where $\mathbf{R}(t)$ now denotes the current position vector of the atom and $\mathbf{V}=\dot{\mathbf{R}}$. The effective detuning $\Delta_{klp}(\mathbf{R}, \mathbf{V})$ is now both position and velocity dependent

$$\Delta_{klp}(\mathbf{R}, \mathbf{V}) = \Delta_0 - \mathbf{V} \cdot \nabla\Theta_{klp}(\mathbf{R}, \mathbf{V}). \quad (36)$$

The dissipative force, proportional to the phase gradient, is given by Eq. (34). This force can be visualized as arising locally from the absorption followed by spontaneous emission of light by the atom. The dipole force, which is proportional to the gradient of the field intensity, subsumed in the position-dependent Rabi frequency, is given by Eq. (35). Both forces play important roles in the cooling and trapping of the atom. The dissipative component is responsible for the existence of a frictional force in a configuration involving two counterpropagating waves, while the dipole force confines the atom to the high-intensity regions of the field when the detuning is below resonance [7].

B. Low-velocity limit

In order to elucidate the nature of the interaction between the LG beam and the atom we consider the low-velocity limit of the dissipative and dipole forces. However, in the computational evaluation of the full extent of the interaction to study the dynamics of the atom described later in this paper, this approximation will not be made. The assumption involved in the low-velocity limit is that the Doppler shift induced by the motion of the atom is smaller than the atomic width $\mathbf{V} \cdot \nabla\Theta \ll \Gamma$. In this case we may expand the denominators of Eqs. (34) and (35) retaining terms up to those linear in the velocity. We can thus write each force as the sum of a static (velocity-independent) and dynamic (velocity-dependent) components. The static components are given by

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}^0(\mathbf{R}) \rangle_{klp} = \frac{2\hbar\Gamma\Omega_{klp}^2(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \left[\{ \eta_{klp}(\mathbf{R}) + k \} \hat{\mathbf{z}} + \frac{l}{r} \hat{\phi} \right. \\ \left. + \xi_k(\mathbf{R}) \hat{\mathbf{r}} \right], \end{aligned} \quad (37)$$

$$\langle \mathbf{F}_{\text{react}}^0(\mathbf{R}) \rangle_{klp} = - \frac{2\hbar\Delta_0\Omega_{klp}(\mathbf{R})\nabla\Omega_{klp}(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \quad (38)$$

and the dynamic components by

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}^V(\mathbf{R}, \mathbf{V}) \rangle = & \frac{4\hbar\Gamma\Delta_0\Omega_{klp}^2(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} \left[\{ \eta_{klp}(\mathbf{R}) + k \} \hat{\mathbf{z}} \right. \\ & + \frac{l}{r} \hat{\phi} + \xi_k(\mathbf{R}) \hat{\mathbf{r}} \left. \right] \left[\{ \eta_{klp}(\mathbf{R}) + k \} V_z + \frac{l}{r} V_\phi \right. \\ & \left. + \xi_k(\mathbf{R}) V_r \right], \end{aligned} \quad (39)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}^V(\mathbf{R}, \mathbf{V}) \rangle = & \frac{2\hbar\Omega_{klp}(\mathbf{R})\nabla\Omega(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]} \\ & \times \left(1 - \frac{2\Delta_0^2}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]} \right) \\ & \times \left[\{ \eta_{klp}(\mathbf{R}) + k \} V_z + \frac{l}{r} V_\phi + \xi_k(\mathbf{R}) V_r \right]. \end{aligned} \quad (40)$$

In the equations above V_z , V_ϕ , and V_r are, respectively, the axial, azimuthal, and radial components of the velocity and the functions $\eta_{klp}(\mathbf{R})$ and $\xi_k(\mathbf{R})$ are defined by

$$\eta_{\pm klp} = \pm \frac{kr^2}{2(z^2 + z_R^2)} \left[1 - \frac{2z^2}{z^2 + z_R^2} \right] \pm (2p \pm l + 1) \frac{z_R}{z^2 + z_R^2}, \quad (41)$$

$$\xi_{\pm k}(\mathbf{R}) = \pm \frac{krz}{z^2 + z_R^2}. \quad (42)$$

We make the additional assumption that the atom moves in a region of the beam for which $z \ll z_R$ and we can then ignore the z dependence in $\Omega_{klp}(\mathbf{R})$ and set $\eta_{klp}(\mathbf{R})=0$ and $\xi_k(\mathbf{R})=0$. We may also write to a good approximation

$$\begin{aligned} \nabla\Omega_{klp}(\mathbf{R}) \approx & \left\{ \left[\frac{|l|}{r} - \frac{2r}{w_0^2} \right] \Omega_{klp}(\mathbf{R}) \right. \\ & \left. - \frac{2\sqrt{2}p}{w_0} \Omega_{k|l|+1p-1}(\mathbf{R}) \right\} \hat{\mathbf{r}}. \end{aligned} \quad (43)$$

In the low-velocity limit with $z \ll z_R$, the static dissipative and dipole forces become

$$\langle \mathbf{F}_{\text{diss}}^0(\mathbf{R}) \rangle_{klp} \approx \frac{2\hbar\Gamma\Omega_{klp}^2(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \left[k\hat{\mathbf{z}} + \frac{l}{r} \hat{\phi} \right], \quad (44)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}^0(\mathbf{R}) \rangle_{klp} \approx & -\frac{2\hbar\Delta_0\Omega_{klp}(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \left(\left[\frac{|l|}{r} - \frac{2r}{w_0^2} \right] \Omega_{klp}(\mathbf{R}) \right. \\ & \left. - \frac{2\sqrt{2}p}{w_0} \Omega_{k|l|+1p-1}(\mathbf{R}) \right) \hat{\mathbf{r}}, \end{aligned} \quad (45)$$

while the dynamic dissipative and dipole forces become

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}^V(\mathbf{R}, \mathbf{V}) \rangle \approx & \frac{4\hbar\Gamma\Delta_0\Omega_{klp}^2(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} \left[\left\{ k^2 V_z + \frac{kl}{r} V_\phi \right\} \hat{\mathbf{z}} \right. \\ & \left. + \frac{kl}{r} V_z \hat{\phi} \right], \end{aligned} \quad (46)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}^V(\mathbf{R}, \mathbf{V}) \rangle \approx & \frac{2\hbar\Omega_{klp}(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} \\ & \times \left(1 - \frac{2\Delta_0^2}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]} \right) \\ & \times \left[k V_z + \frac{l}{r} V_\phi \right] \left(\left[\frac{|l|}{r} - \frac{2r}{w_0^2} \right] \Omega_{klp}(\mathbf{R}) \right. \\ & \left. - \frac{2\sqrt{2}p}{w_0} \Omega_{k|l|+1p-1}(\mathbf{R}) \right) \hat{\mathbf{r}}. \end{aligned} \quad (47)$$

Equations (44) and (46) show that the dissipative force has static components in both the axial and azimuthal directions; the latter is equivalent to a torque about the beam axis. These forces combine with dynamic components in the axial and azimuthal directions. Note that within this approximation, Eq. (46) shows that there is a reciprocal relationship between the axial and azimuthal motions. An atom moving initially in the z direction will induce a force in the azimuthal direction and vice versa. It may be seen from Eqs. (45) and (47) that the dipole force consists of static and dynamic components, both of which are in the radial direction.

The static component of the dipole force, given by Eq. (38), attracts the atom to the high-intensity regions of the field when the detuning is below resonance. This force can be derived from a potential [7]

$$\langle U(\mathbf{R}) \rangle_{klp} = \frac{\hbar\Delta_0}{2} \ln \left[1 + \frac{2\Omega_{klp}^2(\mathbf{R})}{\Delta_0^2 + \Gamma^2} \right] \quad (48)$$

such that $\langle \mathbf{F}_{klp}^0 \rangle = -\nabla \langle U(\mathbf{R}) \rangle_{klp}$. This potential exhibits minima in the high-intensity regions of the beam for an atom tuned below resonance where $\Delta_0 < 0$. For $\Delta_0 > 0$, we have trapping in the low-intensity (dark) regions of the field. As an illustration, we consider the LG mode for which $l=1$, $p=0$. The potential is

$$\langle U \rangle_{k10} = \frac{\hbar\Delta_0}{2} \ln \left[1 + \frac{2\Omega_{k10}^2(\mathbf{R})}{\Delta_0^2 + \Gamma^2} \right]. \quad (49)$$

At the beam waist $z=0$, the minimum occurs at $r=r_0$ where

$$r_0 = w_0/\sqrt{2}. \quad (50)$$

For a beam propagating along the z axis it is easy to verify that the locus of the potential minimum in the xy plane is a circle given by

$$x^2 + y^2 = r_0^2. \quad (51)$$

Expanding the potential in powers of $(r-r_0)$ we have the parabolic approximation

$$\langle U \rangle_{k10} \approx U_0 + \frac{1}{2} \Lambda_{k10} (r-r_0)^2, \quad (52)$$

where U_0 is the potential depth given by

$$U_0 = \frac{1}{2} \hbar \Delta_0 \ln \left[1 + \frac{2\Omega_{k10}^2(r_0)}{\Delta_0^2 + \Gamma^2} \right] \quad (53)$$

and Λ_{k10} is an effective elastic constant given by

$$\Lambda_{k10} = \frac{4\hbar|\Delta_0|}{\Delta_0^2 + 2e^{-1}\Omega_{k00}^2 + \Gamma^2} \left(\frac{e^{-1}\Omega_{k00}^2}{w_0^2} \right). \quad (54)$$

The atom is considered trapped if its kinetic energy is less than U_0 and will exhibit quasi-harmonic vibrational motion about $r=r_0$. The characteristic angular frequency is equal to $\sqrt{\Lambda_{k10}/M}$, where M is the atomic mass.

C. Counterpropagating LG beams

1. One-dimensional case

We have seen above that an atom immersed in a Laguerre-Gaussian beam will experience a dissipative force that is predominantly in the direction of propagation and a dipole force in the radial direction. If a second beam is added propagating in the opposite direction, we have a configuration that can be referred to as the one-dimensional (1D) counterpropagating beam configuration. In this paper the beams are assumed to be independent of each other in that their phases are not locked. The case in which the beams are phase locked to form a standing wave will be considered elsewhere. For independent counterpropagating LG beams we can write the mean force on the atom as a sum of forces due to individual beams

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}} \rangle_{kl_1p_1, -kl_2p_2} &= 2\hbar\Gamma\Omega_{klp}^2(\mathbf{R}) \\ &\times \left[\frac{\nabla \Theta_{kl_1p_1}(\mathbf{R})}{\Delta_{kl_1p_1}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{kl_1p_1}^2(\mathbf{R}) + \Gamma^2} \right. \\ &\quad \left. + \frac{\nabla \Theta_{-kl_2p_2}(\mathbf{R})}{\Delta_{-kl_2p_2}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{-kl_2p_2}^2(\mathbf{R}) + \Gamma^2} \right], \end{aligned} \quad (55)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}} \rangle_{kl_1p_1, -kl_2p_2} &= -2\hbar\Omega_{klp}(\mathbf{R})\nabla\Omega_{klp} \\ &\times \left[\frac{\Delta_{kl_1p_1}(\mathbf{R}, \mathbf{V})}{\Delta_{kl_1p_1}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{kl_1p_1}^2(\mathbf{R}) + \Gamma^2} \right. \\ &\quad \left. + \frac{\Delta_{-kl_2p_2}(\mathbf{R}, \mathbf{V})}{\Delta_{-kl_2p_2}^2(\mathbf{R}, \mathbf{V}) + 2\Omega_{-kl_2p_2}^2(\mathbf{R}) + \Gamma^2} \right], \end{aligned} \quad (56)$$

where we have assumed that $p_1=p_2=p$ and either $(l_1=-l_2=l)$ or $(l_1=l_2=l)$.

In the low-velocity regime, for an atom close to the beam waist, we may make use of Eqs. (44)–(47). The total static dissipative and dipole forces are then given by

$$\langle \mathbf{F}_{\text{diss}}^0(\mathbf{R}) \rangle_{kl_1p_1, -kl_2p_2} \approx \frac{2\hbar\Gamma\Omega_{klp}^2(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \left[\frac{l_1}{r} + \frac{l_2}{r} \right] \hat{\phi}, \quad (57)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}^0(\mathbf{R}) \rangle_{kl_1p_1, -kl_2p_2} &\approx -\frac{4\hbar\Delta_0\Omega_{klp}(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \\ &\times \left(\left[\frac{|l|}{r} - \frac{2r}{w_0^2} \right] \Omega_{klp}(\mathbf{R}) \right. \\ &\quad \left. - \frac{2\sqrt{2}p}{w_0} \Omega_{k|l|+1, p-1}(\mathbf{R}) \right) \hat{\mathbf{r}} \end{aligned} \quad (58)$$

and the total dynamic dissipative and dipole forces are

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}^V(\mathbf{R}, \mathbf{V}) \rangle_{kl_1p_1, -kl_2p_2} &\approx \frac{4\hbar\Gamma\Delta_0\Omega_{klp}^2(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} \\ &\times \left\{ 2k^2V_z + \frac{k}{r}(l_1 - l_2)V_\phi \right\} \\ &\times \hat{\mathbf{z}} + \frac{k}{r}(l_1 - l_2)V_z\hat{\phi}, \end{aligned} \quad (59)$$

$$\begin{aligned} \langle \mathbf{F}_{\text{dipole}}^V(\mathbf{R}, \mathbf{V}) \rangle_{kl_1p_1, -kl_2p_2} &\approx \frac{2\hbar\Omega_{klp}(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]} \\ &\times \left(1 - \frac{2\Delta_0^2}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]} \right) \\ &\times \left(\left[\frac{|l|}{r} - \frac{2r}{w_0^2} \right] \Omega_{klp}(\mathbf{R}) \right. \\ &\quad \left. - \frac{2\sqrt{2}p}{w_0} \Omega_{k|l|+1, p-1}(\mathbf{R}) \right) \\ &\times (l_1 + l_2) \frac{V_\phi}{r} \hat{\mathbf{r}}. \end{aligned} \quad (60)$$

From Eq. (58) we see that the velocity-independent dipole force is simply double that of a single-beam case. The dissipative force, however, depends on the relative signs of l_1 and l_2 . For $l_1=l_2=l$ we have

$$\langle \mathbf{F}_{\text{dipole}}^0(\mathbf{R}) \rangle_{klp-k-lp} \approx \frac{4\hbar\Gamma\Omega_{klp}^2(\mathbf{R})}{\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2} \left(\frac{l}{r} \right) \hat{\phi}, \quad (61)$$

$$\langle \mathbf{F}_{\text{diss}}^V(\mathbf{R}, \mathbf{V}) \rangle_{klp-k-lp} \approx \frac{8\hbar\Gamma\Delta_0\Omega_{klp}^2(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} k^2V_z\hat{\mathbf{z}}. \quad (62)$$

Thus, for $l_1=l_2=l$ we have a torque about the beam axis and an axial cooling or heating force, depending on the sign of Δ_0 .

For the case $l_1=-l_2=l$, we have

$$\langle \mathbf{F}_{\text{diss}}^0(\mathbf{R}) \rangle_{klp-k-lp} = 0, \quad (63)$$

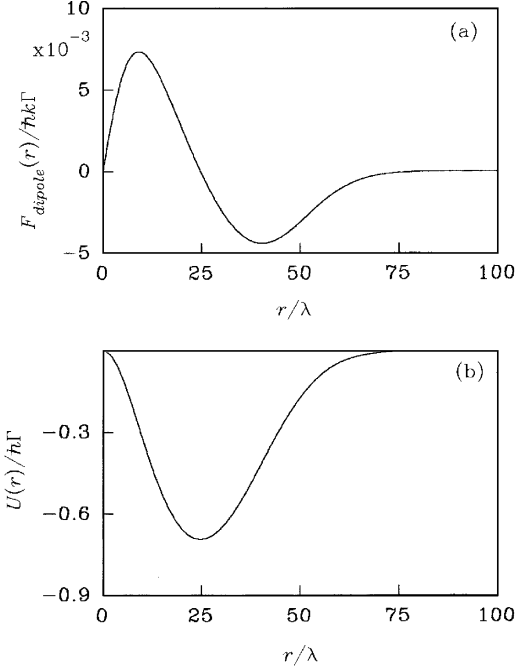


FIG. 2. (a) Radial distribution of the dipole force due to 1D counterpropagating Laguerre-Gaussian beams at $z=0$. Here $l_1 = -l_2 = 1$, $p_1 = p_2 = 0$, and the parameters are $\Delta_0 = -\Gamma$, $\Omega_{k00} = 1.648\Gamma$, and $\omega_0 = 35\lambda$. (b) Radial potential distribution corresponding to (a).

$$\begin{aligned} \langle \mathbf{F}_{\text{diss}}^V(\mathbf{R}, \mathbf{V}) \rangle_{klp-k-lp} &\approx \frac{8\hbar\Gamma\Delta_0\Omega_{klp}^2(\mathbf{R})}{[\Delta_0^2 + 2\Omega_{klp}^2(\mathbf{R}) + \Gamma^2]^2} \\ &\times \left\{ \left[k^2 V_z + \frac{kl}{r} V_\phi \right] \hat{\mathbf{z}} + \frac{kl}{r} V_z \hat{\phi} \right\}. \end{aligned} \quad (64)$$

The static force in this case is zero, while the velocity-dependent force contains extra terms that arise from the orbital angular momentum of the counterpropagating Laguerre-Gaussian beams. As with the one-beam case, we again see force components arising from the reciprocating interchange between the axial and azimuthal motions.

2. Two-dimensional, three-dimensional, and three coplanar beams

The 2D case arises when a second pair of counterpropagating LG beams is arranged orthogonal to the first pair. The total force can again be written as a sum of forces from each of the beams. However, in addition to the reciprocal action between the azimuthal and axial motions in each pair of beams, there is also the fact that the azimuthal atomic motion associated with one beam is part of the axial motion in the other. In other words, there is an additional level of reciprocity between the components of the motion arising from the presence of two pairs of counterpropagating beams.

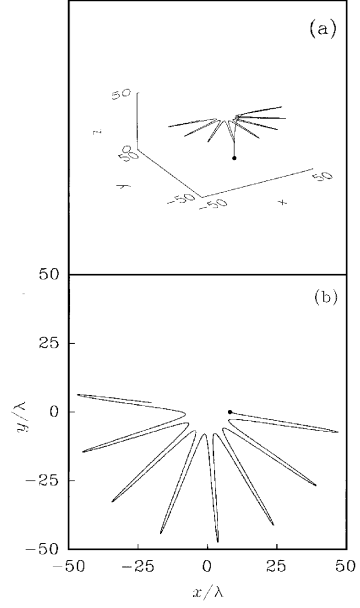


FIG. 3. (a) Trajectory of a Mg^+ ion in 1D counterpropagating LG beams for $l_1 = -l_2 = 1$ and $p_1 = p_2 = 0$. All distances are in units of the wavelength of the light λ . The initial position is $\mathbf{R}_0 = 8\lambda\hat{\mathbf{x}}$ and the initial velocity components are $V_z = 5.0 \text{ ms}^{-1}$ and $V_r = 0 = V_\phi$. (b) Projection in the xy plane of the ion trajectory shown in (a). In this and subsequent figures, the initial position is indicated by a full circle. See the text for the values assumed for the other parameters.

There are also two overlapping dipole potential distributions arising from the orthogonal beams. It is easy to see for beam pairs for which $l_1 = -l_2 = 1$ and $p_1 = p_2 = 0$ and where the axes are such that one pair is along the z axis and the second along the x axis, the potential minima are four times as deep as that of a single beam. The minima are situated at the space points defined by the two equations

$$x^2 + y^2 = w_0^2/2, \quad (65)$$

$$y^2 + z^2 = w_0^2/2. \quad (66)$$

These equations apply the additional constraint $x = \pm z$. Atoms subject to such 2D counterpropagating beams will conglomerate at points lying on the curve defined by two intersecting circles, one on the plane $x+z=0$ and the other on the plane $x-z=0$.

When a third set of beams is arranged orthogonal to the other two orthogonal pairs we have 3D counterpropagating LG beams. The common potential minima in this case occur at eight distinct points defined by

$$x = \pm w_0/\sqrt{2}, \quad y = \pm w_0/\sqrt{2}, \quad z = \pm w_0/\sqrt{2} \quad (67)$$

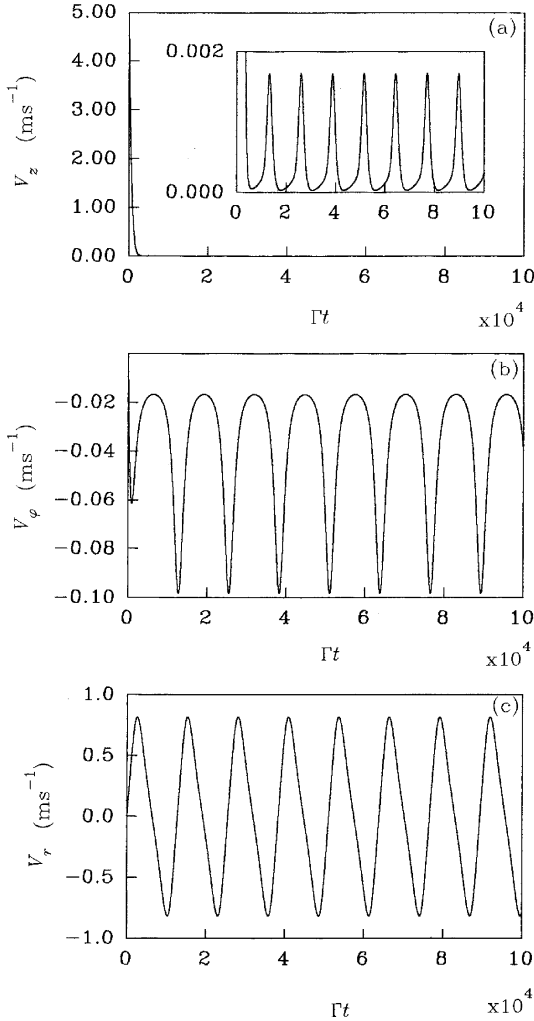


FIG. 4. Variations of the velocity components for the case in Fig. 3. (a) Evolution of V_z indicating axial cooling. The inset to this figure shows small oscillations of V_z due to reciprocating effects. (b) Evolution of V_ϕ and (c) evolution of V_r . Both (b) and (c) indicate the rapid onset of oscillatory motions of the same period.

and are six times as deep as the potential due to one beam. However, in this case the detailed polarization gradients are such as to make further study of this configuration nontrivial.

Finally, we consider the case of three coplanar beams [19] in the x - y plane in a symmetric configuration in which the angle between adjacent beams is $2\pi/3$. This leads to three overlapping circles that meet at two distinct points at

$$x=0, \quad y=0, \quad z=\pm w_0/\sqrt{2} \quad (68)$$

and the potential well is three times as deep as for a single beam.

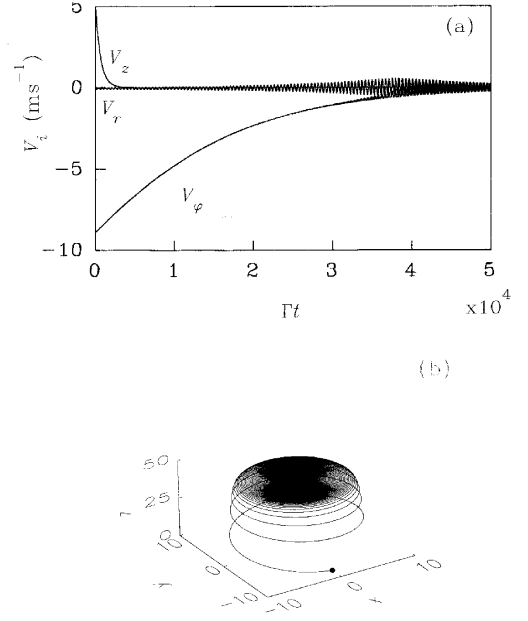


FIG. 5. (a) Evolution of the velocity components of a Mg^+ ion subject to Laguerre-Gaussian 1D counterpropagating beams with $l_1=l_2=1$, $p_1=p_2=0$, and $|\mathbf{B}|=1$ T. Initially the ion possesses both azimuthal and axial velocity components $V_z=5.0$ ms^{-1} and $V_\phi=-8.9$ ms^{-1} . (b) Trajectory of the Mg^+ in (a). All distances are in units of the wavelength of the light λ .

IV. Mg^+ IN MULTIPLE BEAMS

The emphasis throughout this paper is on the physics introduced by the orbital angular momentum aspects of the interaction of atoms with Laguerre-Gaussian light. In the theoretical analysis presented in the preceding section we were able to infer that an atom in a configuration of such beams is subject to axial forces and various forms of static and dynamic rotational forces and that axial and rotational motions influence each other in a rather intricate way. Furthermore, a system of multiple Laguerre-Gaussian beams presents an atom with well-defined potential landscapes that depend on the angular momentum quantum numbers and beam configuration. For example, in the 1D case with a given set of parameters, a given atom should have well-defined quasi-harmonic vibrational states associated with the potential profiles.

To illustrate these features we consider the case of Mg^+ in Laguerre-Gaussian light. The Mg^+ mass is $M=4.0 \times 10^{-26}$ kg; the transition wavelength is $\lambda=280.1$ nm and its half-width is $\Gamma=2.7 \times 10^8$ s^{-1} . To illustrate the theory typical beam parameters are exemplified by the choices $\Delta_0=-\Gamma$, $\Omega_{k00}=1.648\Gamma$, and $w_0=35\lambda$. The equation of motion of a Mg^+ ion in multiple LG beams is written as

$$M \frac{\partial^2 \mathbf{R}(t)}{\partial t^2} = \sum_j \{ \langle \mathbf{F}_{\text{diss}}(\mathbf{R}, \mathbf{V}) \rangle_{k_i l_i p_i} + \langle \mathbf{F}_{\text{dipole}}(\mathbf{R}, \mathbf{V}) \rangle_{k_i l_i p_i} \} + Q \mathbf{V} \times \mathbf{B}, \quad (69)$$

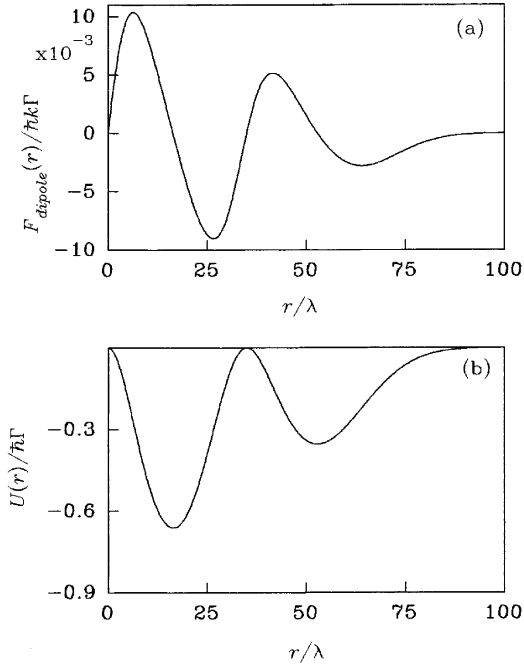


FIG. 6. (a) Radial distribution of the dipole force due to 1D counterpropagating Laguerre-Gaussian beams at $z=0$. Here $l_1=-l_2=1$, $p_1=p_2=1$, and the parameters are $\Delta_0=-\Gamma$, $\Omega_{k00}=1.648\Gamma$, and $\omega_0=35\lambda$. (b) Radial potential distribution corresponding to (a).

where Q is the ionic charge and we have included the last term on the right-hand side to allow for the possibility of an applied external magnetic field. The summation indicates the vector addition over force contributions arising from individual beams. The forces from each beam are taken in their unapproximated forms given by Eqs. (34) and (35).

We begin by considering 1D counterpropagating beams in the absence of the magnetic field. Figure 2(a) displays the dipole force as given by Eq. (56) as a function of radial distance r at $y=0$ and $z=0$. The beam quantum numbers are such that $l_1=-l_2=1$ and $p_1=p_2=0$ and the parameters are $\Delta_0=-\Gamma$, $\Omega_{k00}=1.648\Gamma$, and $w_0=35\lambda$. Figure 2(b) displays the corresponding radial potential distribution. The maximum intensity is located at points where $r=w_0/\sqrt{2}=24.75\lambda$. As expected, we see that for $\Delta_0<0$ the dipole potential exhibits a minimum at points where the intensity is maximum. The vibrational states in the parabolic approximation have an elastic constant that is twice that for the one-beam case as given by Eq. (54). The vibrational frequency corresponding to the above parameters is

$$\nu \approx \frac{1}{2\pi} \left(\frac{8\hbar|\Delta_0|e^{-1}\Omega_{k00}^2}{Mw_0^2[\Delta_0^2+2e^{-1}\Omega_{k00}^2+\Gamma^2]} \right)^{1/2}. \quad (70)$$

For the parameter values specified above this yields $\nu \approx 2.0 \times 10^4 \Gamma$.

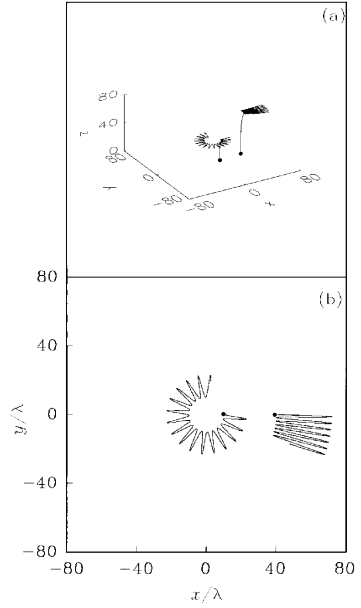


FIG. 7. (a) Trajectories of a Mg^+ ion in 1D counterpropagating LG beams for $l_1=-l_2=1$ and $p_1=p_2=1$ for two different initial positions: one is at $\mathbf{R}_0=10\lambda\hat{x}$ and the second is at $\mathbf{R}_0=40\lambda\hat{x}$. All distances are in units of the wavelength of the light λ . The initial velocity components in both cases are $V_z=5.0 \text{ ms}^{-1}$ and $V_r=0=V_\phi$. (b) Projections in the xy plane of the ion trajectories shown in (a). See the text for the assumed values of the other parameters.

Figure 3(a) displays the trajectory of the ion as a function of time and Fig. 3(b) depicts its projection onto the xy plane. The initial position is at $\mathbf{R}_0=8\lambda\hat{x}$ and the initial velocity components are $V_z=5.0 \text{ ms}^{-1}$, $V_\phi=0=V_r$. It is clear from the figure that the atom, subject to an axial friction force, has been slowed axially. Once the atom is moving sufficiently slowly, it starts a vibrational motion about the radial coordinate $r=w_0/\sqrt{2}$, accompanied by a slow rotational motion. The latter, according to Eq. (46), is attributed to the azimuthal component of the dissipative force induced by the axial motion.

Figure 4 displays the evolution of the velocity components. The axial velocity is seen to decay almost to zero. However, closer inspection, as shown by the inset to Fig. 4(a), reveals that the axial motion exhibits periodic oscillations that are attributable to a reciprocating force arising from the periodic azimuthal motion, depicted in Fig. 4(b). The period associated with these figures is indeed about $2.0 \times 10^4 \Gamma$ as in the estimate based on Eq. (70). An important feature displayed by the results depicted in Figs. 3 and 4 is that changing the sign of the angular momentum quantum number l from $+1$ to -1 , which is readily achievable [8,9], causes the change in the rotational motion from clockwise to the opposite (counterclockwise) sense.

Figure 5 is concerned with the case $l_1=l_2=+1$ and $p_1=p_2=0$ in the presence of a magnetic field $|\mathbf{B}|=1 \text{ T}$ directed along the positive z axis. From Eqs. (61) and (62) we

deduce that, besides the ion cyclotron motion due the magnetic field, the main effects are in the form of an axial friction force provided that $\Delta_0 < 0$ and a static torque about the beam axis that acts upon the ion azimuthally. Figure 5(a) displays the evolution of velocity components for the case $\Delta_0 = -\Gamma$. The initial ion position is $\mathbf{R}(0) = -8\lambda\hat{\mathbf{y}}$ and the initial velocity components are $V_r = 0$, $V_\phi = -8.9 \text{ ms}^{-1}$, and $V_z = 5.0 \text{ ms}^{-1}$. We see that the torque due to the LG beams generates a braking effect on the cyclotron motion, while the axial motion is gradually cooled by the axial friction force. All these features can be inferred from the trajectory shown in Fig. 5(b). If the sign of l in both beams were to be changed, but $\mathbf{B}$ kept in the same direction, we would have heating of the azimuthal motion, while the axial motion would still be cooled. Clearly the former case amounts to a decrease in angular motion due to the LG beam, while the latter is equivalent to the enhancement of the angular motion. These phenomena are attributable only to the angular momentum properties of the LG beams [20].

Figure 6 is concerned with counterpropagating beams with the next-higher-order LG modes $l_1 = -l_2 = 1$ and $p_1 = p_2 = 1$ and no external magnetic field. Other parameter values are $\Omega_{k00} = 1.648$ and $\Delta_0 = -\Gamma$. Figures 3(a) and 3(b) display the radial distribution of the dipole force and corresponding potential, respectively. We now have two potential wells with minima at $r = 0.468w_0 = 16.38\lambda$ and $r = 1.5w_0 = 52.86\lambda$. The ion is destined to oscillate about one of these points, depending on the initial conditions. This can be seen in Fig. 7 for an atom with $V_x = 0 = V_y$ and $V_z = 5.0 \text{ ms}^{-1}$. The inner curve depicts the trajectory when the atom begins at $\mathbf{R}_0 = 10\lambda\hat{\mathbf{x}}$ and the outer curve when it begins $\mathbf{R}_0 = 40\lambda\hat{\mathbf{y}}$.

Figure 8(a) shows the trajectory in a 2D counterpropagating beam case with $l_1 = -l_2 = 1$ and $p_1 = p_2 = 0$ and Fig. 8(b) shows the corresponding projection in the xy plane. The initial position is at $\mathbf{R}_0 = 10\lambda\hat{\mathbf{x}}$ and the initial velocity components are $(0.02, 0.02, 0.05) \text{ ms}^{-1}$. Figure 9 shows the evolution of the velocity components. From Figs. 8 and 9 it can be seen that the atom is subject to friction forces from all directions, which result in it coming to rest at a point within the potential profile. We have shown earlier that the locus of the dipole potential minimum for the 2D case with $(1,0)$ beams is in the form of two intersecting circles, satisfying Eqs. (65) and (66). This is shown in Fig. 10 for the case $w_0 = 35\lambda$. The trajectory end point for the case depicted in Figs. 8 and 9 lies on the curve shown in Fig. 10. Thus this theory assigns predetermined end points for a given ion under given initial conditions. That this is clearly the case can be seen from Table I, where the coordinates of the trajectory end point recorded for various starting points satisfy Eqs. (65) and (66) for the two intersecting circles shown in Fig. 10.

V. COMMENTS AND CONCLUSIONS

In this paper we have explored the nature of the radiation forces and their influence on atomic motion for a specific type of laser light, namely, Laguerre-Gaussian laser light in the form of a single beam and for multiple beams in various configurations. We have emphasized from the outset that the orbital angular momentum effects characterizing these modes give rise to different physical phenomena when such

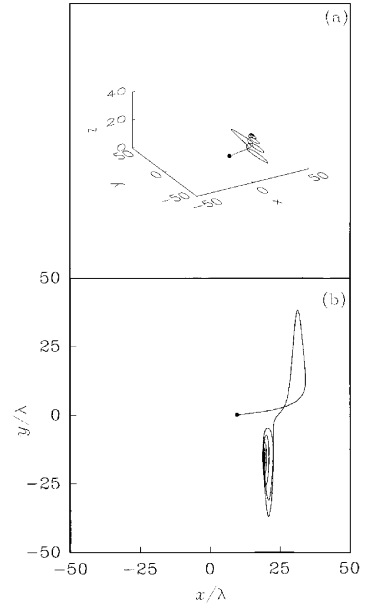


FIG. 8. (a) Trajectory of a Mg^+ ion in 2D counterpropagating LG beams involving two orthogonal pairs: one pair has the z axis as a common axis with $l_1 = -l_2 = 1$ and $p_1 = p_2 = 0$ and the second pair has the y axis as a common axis and $l_3 = -l_4 = 1$ and $p_3 = p_4 = 0$. All distances are in units of the wavelength of the light λ . The initial position of the Mg^+ ion is at $\mathbf{R}_0 = 20\lambda\hat{\mathbf{x}}$ and the initial velocity components are $(0.02, 0.02, 0.05) \text{ ms}^{-1}$. (b) Projection in the xy plane of the trajectory in (a).

light is made to interact with atoms at near resonance. We have shown that a variety of forces come into play when the LG light is arranged in well-defined multiple beams particularly linear, orthogonal 2D, and symmetric coplanar three-beam configurations. We have, for simplicity, considered only coaxial multiple LG beams of the same kind whose beam waists coincide and have assumed that all counterpropagating pairs have the same magnitude of orbital angular momentum quantum numbers l and p . Notwithstanding the simplification inherent in these symmetric configurations, the physics has been intricate, but has given rise to effects associated with the orbital angular momentum of LG beams.

The results show that LG light generates a potential arising from the dipole force, while the dissipative force provides a mechanism to cool the atom axially and that there is a torque that can be utilized to cool or heat the azimuthal motion. Furthermore, there are reciprocating forces involving an interplay between motions in orthogonal directions that can generate oscillatory and precessional motions.

The model we have adopted involves linearly polarized light. We have also assumed that the beams are independent and possess no fixed phase relationship, thus excluding interference or multiphoton processes [1], for example, absorption from one beam followed by emission into the other beam. The atom responds therefore to the sum of the individual forces acting upon it. This is distinct from the case in which the two counterpropagating beams form a standing

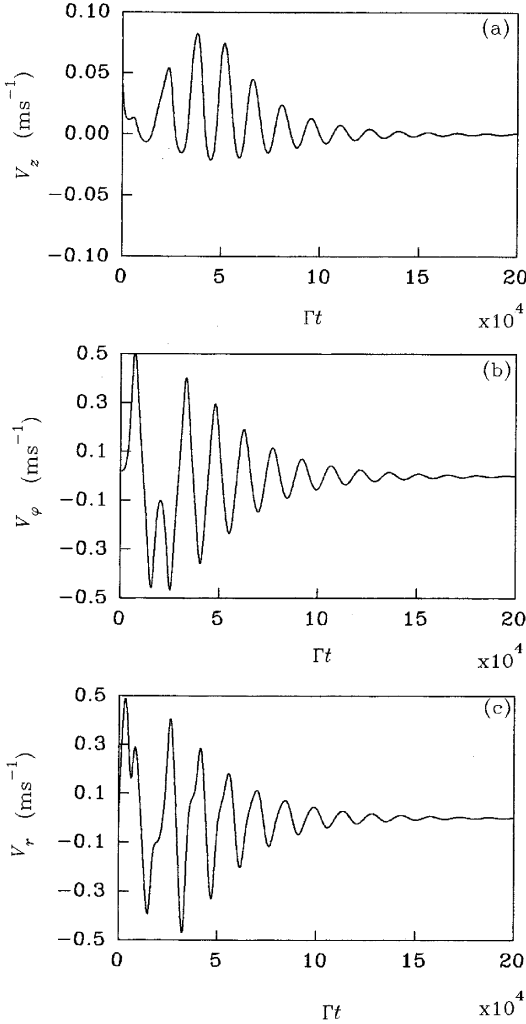


FIG. 9. Evolution of the velocity components corresponding to case with initial position at $\mathbf{R}_0 = 10\lambda\hat{\mathbf{x}}$ (a) $V_z(t)$, (b) $V_\phi(t)$, and (c) $V_r(t)$. Note that all components of velocity go to zero after a sufficiently long time.

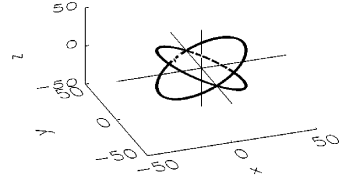


FIG. 10. Locus of spatial points where the dipole potential profile due to a system of two orthogonal pairs of counterpropagating LG beams possesses the lowest minimum. All distances are in units of the wavelength of the light λ .

wave and the possible Doppleron effects [1] that can arise under such circumstances. Work on the case of a standing LG wave is planned to be reported elsewhere.

In summary, this paper has dealt with the basic features that can arise when an atom interacts with multiple Laguerre-Gaussian beams possessing orbital angular momentum. The main effects of LG light on atomic behavior are elucidated for the 1D counterpropagating beams where we find reciprocity between axial and azimuthal motions and the existence of a static torque and a characteristic dipole potential. In the case of 2D counterpropagating beams the potential profile indicates that cooled atoms are forced to congregate into well-defined loci depending on the size of the light beams. Initial cooling need not be effected by the same LG beams; the primary aim of the LG beams at the late cooling stages could be the installation of the dipole potential; artificially generated dark field beams have been experimentally exploited [21]. We have illustrated the results by considering the case of beams of order of tens of wavelengths diameter. This results in atoms sitting on loci separated by distances of tens of wavelengths. However, the generation of results for diameters of order of millimeters is straightforward and would lead to atomic loci separated by distances in the mms scale. We have also briefly considered the loci for the 3D case and for the three coplanar converging LG beams. For these cases no solutions of the dynamical equation were presented and we have only pointed out the characteristic potential profiles and the points at which cooled atoms would congregate.

The effects of orbital angular momentum have been discussed here in connection with Laguerre-Gaussian modes. It

TABLE I. Coordinates x_f, y_f, z_f of the trajectory end points against initial coordinates x_0, y_0, z_0 . All distances are in units of the wavelength of the light λ . The last three columns demonstrate that the ion end points always lie on the two intersecting circles shown in Fig. 10.

x_0	y_0	z_0	x_f	y_f	z_f	$\sqrt{x_f^2 + y_f^2}$	$\sqrt{y_f^2 + z_f^2}$	$w_0/\sqrt{2}$
20	0	0	15.96	18.92	15.96	24.75	24.75	24.75
20	20	0	2.84	24.61	2.84	24.75	24.75	24.75
-20	20	10	-14.27	20.22	14.27	24.75	24.75	24.75
-20	20	-10	-13.81	20.54	-13.81	24.75	24.75	24.75

appears likely, however, that the dynamically induced stabilization of the atomic motion, the so-called supermolasses configuration, which arises from a small displacement of the molasses fields [22], can be related to the azimuthal forces arising from orbital angular momentum, as probably can the macroscopic vortex force due to the offset beams in the spin-polarized spontaneous force-atom trap of Walker *et al.* [23]. As we have shown, the orbital angular momentum of the Laguerre-Gaussian modes is explicit and their influence on

atomic motion is more straightforward to interpret.

ACKNOWLEDGMENTS

The authors are grateful to Dr. E. Riis and Dr. R. C. Thompson for useful discussions. This work was carried out under EPSRC Grant No. GR/J64009, which also provided support for W.K.L. and V.E.L. L.A. was working at JILA, University of Colorado during the preparation of the paper.

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Selection rules and centre-of-mass motion of ultracold atoms

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Received 3 March 1994, in final form 18 April 1994

Abstract. In recent years much attention has been paid to the quantized evolution of the centre-of-mass momentum and position of ultracold atoms in light fields. We consider the effects resulting from the quantization of the external angular momentum variables. We investigate how spin and orbital angular momentum of light are transferred to internal and external angular momentum of an atom in dipole and quadrupole transitions.

1. Introduction

When a light field interacts with an atom, one can usually neglect the centre-of-mass motion of the atom, and consider the effects on the internal electronic motion only. It suffices, e.g., to specify the change in energy and angular momentum of the electron in the centre-of-mass system.

For a cold atom, however, the recoil effect of even a single photon can no longer be neglected: a single photon can change the external state of the atom. Even in this case the external atomic motion can in general still be described classically. Only when the atom is cooled down further to the recoil temperature, and its de Broglie wavelength λ_B becomes comparable to the wavelength λ of the light, must the external motion be quantized [1].

The prime example of a pure quantum effect arising from the quantized external motion of an ultracold atom is the effect of velocity-selective coherent population trapping [2]. An atom in a light field gets trapped in a state that is insensitive to the light, and which is a superposition of two states with different external momenta, differing by two photon momenta. A cooling scheme based on this mechanism has been shown to lead to temperatures below the recoil temperature [2]. This idea has been extended to three dimensions [3] and to different atomic transitions [4].

Another example is the quantized motion of cold atoms in optical potential wells with the size of a wavelength, produced by two counterpropagating laser beams [5]. Here atoms can be trapped in a single well and can occupy a single quantized energy level therein. Also the occurrence of tunnelling from one well to an adjacent one has been predicted [6]. The presence of the discrete energy levels and the localization of an atom in one well have been observed experimentally [7, 8]. Recently cooling of atoms and their quantized motion in optical wells has also been observed in two- and three-dimensional laser beam configurations [9–11].

Now photons also carry angular momentum [12]. Therefore, photons can exert, in addition to a force, also a torque on an atom. There is one important difference, however:

the momentum of the photon is always transferred to the *external* momentum of the atom, whereas the photon angular momentum in general is transferred to the *internal* atomic angular momentum. In fact, the well known dipole selection rules $\delta l = \pm 1$ refer exclusively to the internal electronic state, and are understood as arising from the spin angular momentum of the photon, which is equal to 1. Recently it has been shown that also orbital angular momentum of light, arising from the transverse spatial dependence of a light beam, is a meaningful concept, accessible to experimental verification [13–16]. This quantity will in general be transferred to the *external* angular momentum of an atom [17]. This implies that it is the *orbital* angular momentum of light that can make atoms rotate around a given point or axis.

Some questions remain: what effects arise for ultracold atoms, when the external angular momentum must be quantized? What happens in a quadrupole transition, where $\delta l = \pm 2$? Does the second unit of angular momentum come from the orbital angular momentum of the photon? In order to answer these questions, we investigate the selection rules concerning (internal and external) angular momenta during the absorption or emission of a photon by an atom. We wish to clarify the conservation laws underlying these rules, and in particular the role of the centre-of-mass motion of the atom. We will discuss some unfamiliar effects on cold atoms resulting from angular momentum transfer between photons and atoms.

Throughout this paper we will use the long-wavelength approximation, i.e. we assume that the wavelength λ of the light is large compared to the size a of the atomic system under consideration. This justifies making an expansion in the small parameter a/λ . It should be noted that this approximation does not imply any restriction on the de Broglie wavelength of the atom: λ_B and a are independent atomic quantities, determined by the size of the external and of the internal part of the wavefunction, respectively. This paper is organized as follows. In order to be able to discuss conservation laws of angular momentum one needs to define electromagnetic field modes with well defined angular momenta. Several possible definitions are reviewed in section 2. In section 3 we examine in some detail the selection rules for photon absorptions and emissions by an atom. Emphasis is put on the modifications that arise due to the inclusion of the centre-of-mass motion of the atom. Some explicit examples are presented in section 4. They serve to show how spin and orbital angular momenta of a photon are distributed over the internal and external angular momenta of the atom, and why the temperature plays an important role. The results are summarized and discussed in section 5. Finally, in the appendix we define the internal and external variables as used in this paper.

2. Photons and angular momentum

We first establish the notation and definitions for the well known multipole waves [12, 18], in which photons are in eigenstates of the operators for total field angular momentum J^2 and its projection J_z . In the second subsection, we define photons in eigenstates of the projected ‘orbital’ and ‘spin’ angular momenta L_z and S_z [17]. Finally, we will briefly discuss Laguerre–Gaussian modes, which have been shown to be producible by a transformation of laser beams [15], and which also possess well defined S_z and L_z .

A photon is here defined as an elementary excitation of a field mode. A field mode is represented by a mode function $F_\lambda(\mathbf{r})$, which is a transverse vector solution to the wave equation with wave number $k_\lambda = \omega_\lambda/c$. Each mode function can be chosen as the eigenfunction of a prescribed set of four commuting Hermitian operators. The mode is then

Table 1. Notation for the quantum numbers related to the various angular momenta of photons and atoms.

Species	Operator	Quantum number	Eigenvalue
Photon	J^2	j	$\hbar^2 j(j+1)$
Photon	J_z	m	$m\hbar$
Photon	S_z	s	$s\hbar k_z/k$
Atom	J_{int}^2	l	$\hbar^2 l(l+1)$
Atom	$J_{\text{int},z}$	m	$m\hbar$
Atom	J_{ext}^2	L	$\hbar^2 L(L+1)$
Atom	$J_{\text{ext},z}$	M	$M\hbar$
Atom	J_{total}^2	J	$\hbar^2 J(J+1)$

specified by the four eigenvalues of these operators, denoted collectively by λ , which are at the same time the quantum numbers of the photons from that field mode.

Once we have thus defined a complete set of normalized transverse vector functions F_λ , the operator A for the vector potential in the Coulomb gauge can be expanded in this set as [19]

$$A(\mathbf{r}) = \sum_{\lambda} \sqrt{\frac{\hbar}{2\epsilon_0\omega_{\lambda}}} [a_{\lambda} F_{\lambda}(\mathbf{r}) + a_{\lambda}^{\dagger} F_{\lambda}^*(\mathbf{r})] \quad (1)$$

where a_{λ} and $a_{\lambda}^{\dagger}$ are the annihilation and creation operators for photons in the mode λ . In the examples we will consider, two of the four quantum numbers will refer to angular momentum variables. The notation for these variables is summarized in table 1.

2.1. Multipole waves

We review here some well known results concerning the multipole waves, as described in [12, 19]. One constructs eigenstates of the four commuting Hermitian operators for total angular momentum J^2 , its projection along the z axis J_z , energy, and parity. The set of quantum numbers is correspondingly given by $\lambda = (j, m, \omega, P)$, where the parity P takes the values ± 1 , and where ω is the frequency. There are two types of waves: electric multipole waves with parity $P = (-1)^j$ and magnetic multipole waves with opposite parity $P = (-1)^{j+1}$. The explicit expressions for the mode functions can be found in [12], or, for the corresponding electric and magnetic fields in [19]. The only multipole fields that are non-vanishing in the origin are the electric dipole waves with $j = 1$ and $m = \pm 1, 0$.

2.2. Bessel waves

One can also construct field modes as the eigenfunctions of the commuting set of operators for the projection of the angular momentum J_z , linear momentum P_z , energy and ‘spin’ S_z . The corresponding quantum numbers are $\lambda = m, k_z, \omega, s$, where $\hbar k_z$ is the momentum in the z direction, and where $s = \pm 1$ denotes the polarization or helicity (right or left hand). Only for waves propagating in the z direction, i.e. for waves with $k_z = \pm k$, is the quantum operator S_z a true spin angular momentum operator [17, 20]. In that case it has discrete eigenvalues $\pm \hbar$. Explicit expressions for the corresponding mode functions F_{λ} were derived in [17]. The result takes the form (the $\pm$ refers to the polarization index s)

$$\frac{1}{\sqrt{2}}(F_x + iF_y) = \frac{k_z \mp k}{2k} f(k_t, k_z, m+1)$$

$$\frac{1}{\sqrt{2}}(F_x - iF_y) = \frac{k_z \pm k}{2k} f(k_t, k_z, m - 1)$$

$$F_z = \frac{k_t}{\sqrt{2}k} f(k_t, k_z, m) \quad (2)$$

where the functions $f(k_t, k_z, m)$ are defined in cylindrical coordinates (ρ, z, ϕ) as

$$f(k_t, k_z, m) = J_m(k_t \rho) \exp(ik_z z) \exp(im\phi) / N. \quad (3)$$

Here J_m is the m th-order Bessel function, $\hbar k_t$ is the momentum in the direction perpendicular to the z direction: $k_t^2 = k^2 - k_z^2$, and N is a normalization constant. Note that these modes do not have well defined total angular momentum J^2 , nor a definite parity. The only fields that are non-vanishing in the origin are those with $m = \pm 1, 0$.

These waves generalize so-called Bessel beams, which are proportional to the zeroth-order Bessel functions, and which have been produced recently to moderately high intensity using a specific zone plate configuration [21].

2.3. Laguerre–Gaussian beams

Laguerre–Gaussian (LG) beams are special solutions to the paraxial equation [22]. They belong, therefore, to a class of exact solutions to an approximate equation describing light beams with a well defined propagation direction. The LG modes are cylindrically symmetric beams, with an azimuthal dependence given by $\exp(im\phi)$. They can be viewed as superpositions of Bessel waves with fixed value of m and with different $k_t \ll k$. These beams can be produced from (laser) Hermite–Gaussian beams by using a special configuration of astigmatic cylindrical lenses [13–15]. Thus one has been able to produce modes with up to three units of orbital angular momentum L_z per photon [15].

3. Selection rules

We review here how the selection rules for photon absorption and emission processes arise in the case that the atomic centre of mass motion is neglected. For more details see [18]. Subsequently we include this external motion, and investigate the resulting modifications of the selection rules and conservation laws.

3.1. Centre of mass is neglected

The atom is modelled by a spinless electron with coordinate $\mathbf{r}$ and momentum $\mathbf{p}$, charge $-e$ and mass μ that is bound to the origin by a given potential $V(\mathbf{r})$. The electron is confined by this potential to move in a region $|\mathbf{r}| < a$. We neglect the centre-of-mass motion, effectively assuming the atom to have an infinitely heavy nucleus. The centre-of-mass position $\mathbf{R}$ is not a dynamical variable, but its value is fixed and equal to $\mathbf{R} = 0$.

The electron interacts with an external electromagnetic field, which is represented by the vector potential $\mathbf{A}$ in the Coulomb gauge. The interaction part of the Hamiltonian is, neglecting the A^2 term†, given by

$$H_I = \frac{e}{\mu} \mathbf{A}(\mathbf{r}) \cdot \mathbf{p}. \quad (4)$$

The wavelength of the light is assumed to be long, so that it is legitimate to expand in a/λ . This boils down to expanding the operator $\mathbf{A}(\mathbf{r})$ (i.e. the field modes $\mathbf{F}_\lambda$) around the origin,

$$\mathbf{A}(\mathbf{r}) = \mathbf{A}(0) + (\mathbf{r} \cdot \nabla) \mathbf{A}(0) + \dots \quad (5)$$

† At the low intensities used in laser cooling this is certainly a valid approximation. Furthermore, in lowest order in a/λ , this term contributes only a (small) energy shift of all states.

3.1.1. *Dipole transitions.* The first term of the expansion yields the dipole interaction

$$H_I^{\text{ED}} = \frac{e}{\mu} \mathbf{p} \cdot \mathbf{A}(0). \quad (6)$$

If the matrix element

$$\langle \psi_f | H_I^{\text{ED}} | \psi_i \rangle$$

of this operator between some initial state $|\psi_i\rangle$ and some final state $|\psi_f\rangle$ is non-zero then the corresponding transition is an electric dipole transition. Let the initial atomic state be an eigenstate of total angular momentum J_{int}^2 and of the projection $J_{\text{int},z}$ with eigenvalues $l(l+1)$ and m : $|\psi_i\rangle = |l, m\rangle$. Then a dipole transition to a final state of the form $|\psi_f\rangle = |l', m'\rangle$ is possible if and only if at least one the matrix elements

$$\langle l', m' | \mathbf{p} | l, m \rangle$$

is non-zero. This leads to the well known dipole selection rules [18] for $\delta m \equiv m' - m$ and for $\delta l \equiv l' - l$,

$$\delta m = \pm 1, 0 \quad \delta l = \pm 1. \quad (7)$$

The usual interpretation of these rules is that they express angular momentum conservation, since the photon has spin 1. Indeed, if one makes an expansion of the field in multipole waves, then only one term has a non-vanishing value at the origin, namely the electric dipole field with total angular momentum $j = 1$, and projections $m = \pm 1, 0$. Thus, angular momentum is conserved, as only the electric dipole wave contributes to the dipole interaction (6). Parity is conserved as well, since the parity of an electric dipole photon is odd, while the parity of the electron state changes sign.

3.1.2. *Higher order transitions.* The second term in the expansion (5), of first order in a/λ , yields two different types of interaction terms: first the magnetic dipole Hamiltonian of the form

$$H_I^{\text{MD}} = \frac{e}{\mu} \vec{Q}_{\text{MD}} : (\mathbf{r}\mathbf{p} - \mathbf{p}\mathbf{r}) \quad (8)$$

and second the electric quadrupole interaction

$$H_I^{\text{EQ}} = \frac{e}{\mu} \vec{Q}_{\text{EQ}} : (\mathbf{r}\mathbf{p} + \mathbf{p}\mathbf{r}). \quad (9)$$

We defined here

$$\vec{Q}_{\text{MD}} = \frac{1}{2}([\nabla \mathbf{A}(0)]^t - \nabla \mathbf{A}(0)) \quad \vec{Q}_{\text{EQ}} = \frac{1}{2}([\nabla \mathbf{A}(0)]^t + \nabla \mathbf{A}(0)) \quad (10)$$

where the superscript t indicates the transpose of a tensor: $T_{ij}^t = T_{ji}$. The quantity $\vec{Q}_{\text{MD}}$ is proportional to the magnetic field, $\vec{Q}_{\text{EQ}}$ to the gradient of the electric field, both evaluated at the origin. A magnetic dipole transition is allowed if one of the matrix elements

$$\langle \psi_f | (\mathbf{r}\mathbf{p} - \mathbf{p}\mathbf{r}) | \psi_i \rangle$$

is non-zero. The off-diagonal components of this tensor operator contain the cartesian components of the angular momentum operator, the diagonal part is a pure number. Neither operator can change the value of l , while only L_x and L_y can change m by one unit. Thus the selection rules are found to be [18]

$$\delta m = \pm 1, 0 \quad \delta l = 0. \quad (11)$$

Analogously, one finds the electric quadrupole selection rules [18]

$$\delta m = \pm 2, \pm 1, 0 \quad \delta l = \pm 2, 0. \quad (12)$$

In the multipole expansion the gradient of the field is non-vanishing at the origin for two kinds of waves: the magnetic dipole wave, with angular momentum 1, and the electric quadrupole wave with angular momentum 2, both with even parity. Hence, the selection rules (11) and (12) express, again, conservation of both angular momentum and parity. The usual comment is, that since the spin of the photon is 1, the second unit of angular momentum in a quadrupole transition must come from orbital angular momentum of the light field. In the next subsection we will show that this is not strictly true. Summarizing we note that in the multipolar expansion of the interaction

(i) in each different order of the expansion the atom interacts with a different multipole wave,

(ii) in lowest (electric dipole) order the spatial field dependence is neglected. Each higher order takes higher-order gradients of the field into account,

(iii) each expansion term of the interaction Hamiltonian is invariant under rotations around the origin. Therefore the selection rules express conservation of angular momentum of the corresponding multipole field and the electron,

(iv) external atomic angular momentum is excluded. The internal angular momentum change is determined completely by the angular momentum of the multipole wave.

3.2. Centre of mass is included

In this subsection we consider a bound system consisting of $N + 1$ charges q_i , with masses μ_i , where the total mass is M and the total charge is zero. The interaction Hamiltonian is given by

$$H_I = - \sum_{i=1}^N \frac{q_i \mathbf{A}(\mathbf{r}_i) \cdot \mathbf{p}_i}{\mu_i}. \quad (13)$$

This has to be rewritten in terms of internal and external variables, by using (A3) from the appendix. The former variables are denoted by Greek symbols, the latter by capital ones. We keep the long-wavelength approximation, but allow the mass M to be finite. We thus include the centre-of-mass motion. Therefore, instead of expanding the field around the origin, we now have an expansion of $\mathbf{A}$ around the centre of mass (see appendix),

$$\begin{aligned} \mathbf{A}(\mathbf{r}_0) &= \mathbf{A}(\mathbf{R}) - \left(\sum_{i=1}^N \frac{\mu_i \boldsymbol{\rho}_i}{M} \cdot \nabla \right) \mathbf{A}(\mathbf{R}) + \dots \\ \mathbf{A}(\mathbf{r}_i) &= \mathbf{A}(\mathbf{R}) - \left(\sum_{j=1}^N \frac{\mu_j \boldsymbol{\rho}_j}{M} \cdot \nabla \right) \mathbf{A}(\mathbf{R}) + (\boldsymbol{\rho}_i \cdot \nabla) \mathbf{A}(\mathbf{R}) + \dots \end{aligned} \quad (14)$$

Notice that one does not obtain the multipolar Hamiltonian in this way. In order to do so, one should in addition apply a unitary transformation [19, 23]: this step is not necessary, however, for obtaining the selection rules.

The state vector of the atom is given by the direct product of an internal part $|\psi\rangle$ and an external part $|\Psi\rangle$. Now also the latter part can change during the emission or absorption of a photon, leading to selection rules for external variables.

3.2.1. *Dipole transitions.* To lowest order in a/λ one finds the dipole term

$$H_I^{\text{ED}} = \sum_{i=1}^N \left(\frac{q_0}{\mu_0} - \frac{q_i}{\mu_i} \right) \mathbf{A}(\mathbf{R}) \cdot \boldsymbol{\pi}_i. \quad (15)$$

Note that the *full* spatial dependence of the field has been taken into account, since $\mathbf{R}$ is now a dynamical variable. Each single term in the summation leads to the usual dipole selection rules for the corresponding internal variables m_i and l_i , whereas the other quantum numbers for $j \neq i$ do not change. Therefore, for the internal quantum numbers m_i, l_i and also for $m = \sum m_i$ and for $\sum l_i$ one finds the selection rules (7).

Additional selection rules now exist for the external angular momentum. They are found from the requirement that the matrix elements of $\mathbf{A}(\mathbf{R})$ between the external parts of the atomic initial and final states be non-vanishing. These selection rules depend explicitly on the spatial dependence of the field modes. In particular, they are different for multipole waves and Bessel waves. Explicit examples will be given in section 4.

Hence, the conservation law now refers to the *total* angular momentum, including the external atomic part. This fact implies, for instance, that an electric dipole transition is also possible in a field with angular momentum $j > 1$. This can be understood in two ways. First, the remaining angular momentum can now be absorbed by the external motion of the atom. Second, the field is sampled by the atomic wavefunction, not only at the origin, but around $\mathbf{R}$ with a width determined by the de Broglie wavelength. Therefore, the fact that a higher-order multipole field vanishes at the origin, no longer implies that the transition is forbidden.

3.2.2. *Higher order transitions.* The first-order terms in H_I , which are linear in the gradient of the vector potential, can be grouped according to

$$\begin{aligned} H_I^1 = & \sum_{i=1}^N \sum_{j=1}^N \left(\frac{q_i}{\mu_i} - \frac{q_0}{\mu_0} \right) \left[\frac{\mu_j \rho_j}{M} \cdot \nabla \right] (\mathbf{A}(\mathbf{R}) \cdot \boldsymbol{\pi}_i) - \sum_{i=1}^N \frac{q_i}{\mu_i} [\boldsymbol{\rho}_i \cdot \nabla] (\mathbf{A}(\mathbf{R}) \cdot \boldsymbol{\pi}_i) \\ & - \sum_{i=1}^N \frac{q_i}{M} [\boldsymbol{\rho}_i \cdot \nabla] (\mathbf{A}(\mathbf{R}) \cdot \mathbf{P}). \end{aligned} \quad (16)$$

For an atom the ratio μ_i/M is small, so that the first term is negligible compared to the second. The third term describes in effect a dipole coupling of the external motion with the magnetic field: it is equivalent to the Röntgen term, which has recently been shown to be necessary for a consistent description of the centre-of-mass motion for cooled atoms [23], and which leads to the existence of a geometric phase analogous to the Aharonov-Casher effect [24]. We concentrate on the second line, which can be separated into two terms,

$$\begin{aligned} H_I^{\text{MD}} &= \tilde{Q}_{\text{MD}}(\mathbf{R}) : \sum_{i=1}^N \frac{q_i}{\mu_i} (\boldsymbol{\rho}_i \boldsymbol{\pi}_i - \boldsymbol{\pi}_i \boldsymbol{\rho}_i) \\ H_I^{\text{EQ}} &= \tilde{Q}_{\text{EQ}}(\mathbf{R}) : \sum_{i=1}^N \frac{q_i}{\mu_i} (\boldsymbol{\rho}_i \boldsymbol{\pi}_i + \boldsymbol{\pi}_i \boldsymbol{\rho}_i) \end{aligned} \quad (17)$$

where $\tilde{Q}_{\text{EQ}}$ and $\tilde{Q}_{\text{MD}}$ are given by the same expressions (10) but now evaluated at $\mathbf{R}$. For each internal angular momentum quantum number m_i and l_i one finds the same selection

rules (11) and (12) as before. For the external angular momentum one finds additional selection rules from the requirement that the gradient of $\mathbf{A}$ have non-zero matrix elements between initial and final external atomic states. Again, the latter rules depend on the explicit spatial dependence of the field modes, and will be different from those in a dipole transition. For details see section 4.

We note that the inclusion of the external atomic motion leads to the following modifications in comparison with the results from the preceding subsection.

(i) In each order of the expansion the atom interacts with the total field.

(ii) In each order the full spatial field dependence is included.

(iii) Each expansion term of the interaction Hamiltonian is invariant under rotations around the origin. Therefore the selection rules express conservation of angular momentum of the total field and the atom.

(iv) External atomic angular momentum is included. The change in internal angular momentum is determined not only by the angular momentum of the field but also by the change in external angular momentum.

4. Illustrations

We start this section by having a closer look at dipole transitions in a field with arbitrary angular momentum $j > 1$. Next we consider electric quadrupole transitions, being the lowest-order transitions in which the atomic angular momentum can change by two units, in a field with angular momentum 1.

4.1. Electric dipole transitions

The transition probability for going from a given initial to some final state in a dipole transition is determined by the square of the matrix element

$$\sum_{i=1}^N \langle \Psi_f | \mathbf{A}(\mathbf{R}) | \Psi_i \rangle \cdot \left(\frac{q_0}{\mu_0} - \frac{q_i}{\mu_i} \right) \langle \psi_f | \boldsymbol{\pi}_i | \psi_i \rangle. \quad (18)$$

If one makes the usual approximation of neglecting the centre-of-mass motion, then first of all the variable $\mathbf{R}$ is a fixed position in space. The first part of the matrix element, then, reduces to

$$\mathbf{A}(\mathbf{R}) \langle \Psi_f | \Psi_i \rangle$$

which implies that, indeed, the external state of the atom does not change. This part will contribute only to the total transition rate, but does not change the angular momentum or the parity of $|\Psi_i\rangle$. Furthermore, if the vector potential vanishes at the centre-of-mass position, then the transition is forbidden. Now, when the external motion is quantized, $\mathbf{R}$ becomes a dynamical variable. The matrix element

$$\langle \Psi_f | \mathbf{A}(\mathbf{R}) | \Psi_i \rangle = \int d\mathbf{R} \Psi_f^*(\mathbf{R}) \mathbf{A}(\mathbf{R}) \Psi_i(\mathbf{R}) \quad (19)$$

is non-zero only when the external state changes; the external state can change parity and can absorb angular momentum. If $\mathbf{A}$ vanishes at the *average* centre-of-mass position, it is no longer implied that the corresponding matrix element vanishes. Still, the electric dipole

transition probability in such a field will be negligibly small in general. For example, in an electric multipole wave with angular momentum j , the field around the origin is proportional to [12, 19]

$$|A(r)| \propto \left(\frac{2\pi r}{\lambda}\right)^{j-1} \quad (20)$$

for $r/\lambda \ll 1$. Thus, if the width of the centre-of-mass wavefunctions of the atom is given by $\Delta x \ll \lambda$, the transition probability is proportional to $(2\pi \Delta x/\lambda)^{2j-2}$, which is negligibly small, except for $j = 1$. For ultracold atoms, however, the width of the wavefunctions of $|\Psi_i\rangle$ and $|\Psi_f\rangle$ is appreciable, as it is determined by the small momentum spread Δp , according to $\Delta x > \hbar/\Delta p$. For atoms cooled down to the recoil limit, $\Delta p \approx \hbar/\lambda$, one obtains

$$\Delta x > \frac{\lambda}{2\pi}. \quad (21)$$

Hence, in this case the field is probed over a distance of the order of the wavelength, so that effectively all higher-order gradients of the field will contribute. Thus the electric dipole transition probability becomes finite, for any multipole wave. As an example, consider an electric quadrupole wave with $j = 2$ and projection $m = 2$. If an atom makes an electric dipole transition, then the selection rules for the internal variables are not changed:

$$\delta l = \pm 1 \quad \delta m = -1, 0, 1. \quad (22)$$

Thus the atom absorbs only one unit of angular momentum, although the photon contains two units. The deficit is accounted for by the external angular momentum of the atom. For instance, the spherical components of A are proportional to

$$(A_x + iA_y) \propto \exp(3i\phi) \quad A_z \propto \exp(2i\phi) \quad (A_x - iA_y) \propto \exp(i\phi) \quad (23)$$

so that the selection rules for external quantum number M are

$$\delta M = 3, 2, 1 \quad (24)$$

such that $\delta m + \delta M = 2$. The quantum number J for the length of the total angular momentum $J_{\text{total}} = J_{\text{int}} + J_{\text{ext}}$, changes according to

$$\delta J = \pm 2 \quad (25)$$

which expresses angular momentum conservation. For L , the eigenvalue pertaining to J_{ext}^2 , the selection rules are less strict and follow from the familiar triangle and parity rules

$$L + l \geq J \geq |L - l| \quad L' + l' \geq J' \geq |L' - l'| \quad (26)$$

and $\delta L = L' - L$ is odd.

4.2. Electric quadrupole transitions

The transition probability for electric quadrupole transitions is determined by the square of the matrix elements

$$\sum_{i=1}^N \langle \Psi_f | \tilde{Q}_{EQ}(\mathbf{R}) | \Psi_i \rangle : \langle \psi_f | \frac{q_i}{\mu_i} (\rho_i \pi_i + \pi_i \rho_i) | \psi_i \rangle. \quad (27)$$

The main conclusions from the preceding subsection can, *mutatis mutandis*, be carried over to this case. For instance, in a field where the quadrupole tensor vanishes in the origin, the probability of making an electric quadrupole transition is negligible, unless the atom is very cold. As an explicit example we consider a circularly polarized Bessel beam (or a Laguerre–Gaussian beam) propagating in the positive z direction, such that $k_z \approx k$, with $L_z = 0$ and $S_z = \hbar$. From (2) one sees that there is only one non-vanishing component of the field:

$$F_x - iF_y = \sqrt{2}f(k_t, k_z, 0) \quad (28)$$

where $k_t \ll k_z$. Suppose an atom absorbs a photon from this beam by making an electric quadrupole transition. Since only one spherical component (28) of the field is non-vanishing, one is left with a reduced set of quadrupole selection rules

$$\delta m = 0, 1, 2 \quad (29)$$

for the projections of internal angular momentum on the z axis. From the explicit spatial dependence of (28) one gets the following selection rules for the external angular momentum,

$$\delta M = 1, 0, -1 \quad (30)$$

respectively, such that $\delta m + \delta M = 1$. Therefore, the magnetic quantum number m can change by 2 units of angular momentum, even though the field possesses only one unit of spin. The second unit does not come from orbital angular momentum of the photon, since $L_z = 0$. Rather the external motion of the atom loses one unit of angular momentum in that case: $\delta M = -1$. For completeness we note that in a wave with $L_z = m\hbar$ and $S_z = \hbar$, the selection rule for δm is not affected, while for δM one has now

$$\delta M = m + 1, m, m - 1 \quad (31)$$

respectively. The extra amount of orbital angular momentum of the photon is thus always transferred to the centre-of-mass motion of the atom. The same conclusion holds for dipole transitions [17]. Similarly, in an electric dipole wave with $j = 1$, the quadrupole transition matrix elements lead to the rules

$$\delta l = \pm 2 \quad (32)$$

even though the field possesses only one unit angular momentum. Again, the other unit comes from the external motion of the atom, not from orbital angular momentum of light.

5. Discussion and conclusions

If one assumes the centre of mass of an atom to be at rest at a given point, the selection rules for one-photon transitions express angular momentum and parity conservation: as is well known [12], an electric or magnetic 2^j transition with $\delta l = j$ is allowed if and only if the emitted or absorbed photon has total angular momentum j , and parity $(-1)^j$ or $(-1)^{j+1}$, respectively.

When the external motion of the atom is included, things are different: a multipole transition is possible also in fields with lower angular momentum. In particular, we showed that a quadrupole transition with $\delta l = \pm 2$ and $\delta m = \pm 2$ is allowed both in a field with total angular momentum $j = 1$, and in a field with vanishing orbital angular momentum $L_z = 0$, and spin $S_z = \pm 1$. Conversely, a multipole transition is also allowed in a field with higher angular momentum: for instance, an atom may emit a quadrupole photon in a dipole transition. In all these cases there is no violation of conservation of angular momentum or parity, since the external motion can absorb angular momentum and since the parity of the external wavefunction can change sign.

These changes originate from the fact that the external centre-of-mass position R must be considered a dynamical variable. Consequently, both the transition probability and the selection rules depend on the explicit spatial dependence of the field. The internal selection rules, on the other hand, are independent of the mode structure, and do not depend on the inclusion of external atomic motion, quantized or classical.

When is it relevant to include the centre-of-mass motion? If the atom is cooled to low temperatures, then the angular momentum transfer, i.e. the torque, of one photon is not negligible. Thus for a cooled atom one expects to see above-mentioned deviations from the usual emission and absorption behaviour. Indeed, one may note that the de Broglie wavelength of the atom becomes larger as its temperature becomes lower. Therefore the cooled atom sees a larger part of the field around its centre-of-mass position. Hence, the spatial dependence of the field becomes important, and thereby also the orbital angular momentum of the field. This part of the field angular momentum is in general absorbed by the external motion of the atom, in the sense discussed in section 4.

This also indicates how an atom can be made to rotate around a given axis. When the atom is cooled in a Doppler cooling scheme with two counter-propagating Laguerre-Gaussian laser beams with azimuthal index m (i.e. $L_z = m\hbar$ per photon), then at each stimulated absorption the atomic external angular momentum changes by $m\hbar$. The angular momentum will on average not change by spontaneous emissions. The net rate of absorption of external angular momentum is then equal to $dM/dt = AP_e m\hbar$, where A is the spontaneous decay rate and P_e the probability for the atom to be in the excited state.

Finally, let us note that we discussed only single-photon transitions. Multi-photon transitions can, to lowest order, be seen as a sequence of electric dipole transitions, and the foregoing conclusions can easily be extended to this case.

Acknowledgments

It is a pleasure to thank Professor P Lambropoulos for fruitful discussions. This work is supported by the Human Capital and Mobility programme under contract no ERBCHBGCT920192.

Appendix A. Internal and external variables

We consider an $N + 1$ -particle system, consisting of particles $i = 0 \dots N$. Their positions and momenta are denoted by $\mathbf{r}_i$ and $\mathbf{p}_i$. Here we will define internal and external position and momentum variables for this system. The internal variables will be denoted by greek symbols, the external variables by capital symbols. The internal position variables $\boldsymbol{\rho}_i$ for $i = 1 \dots N$ are chosen relative to the position of particle $i = 0$, which for an atom would be the nucleus. The external position vector $\mathbf{R}$ is the centre-of-mass position. Thus we define

$$\mathbf{R} = \sum_{j=0}^N \frac{\mu_j \mathbf{r}_j}{M} \quad \boldsymbol{\rho}_i = \mathbf{r}_i - \mathbf{r}_0 \quad (\text{A1})$$

for $i = 1 \dots N$, where $M = \sum \mu_i$ is the total mass. The canonically conjugate momenta follow from these definitions, and are given by

$$\mathbf{P} = \sum_{j=0}^N \mathbf{p}_j \quad \boldsymbol{\pi}_i = \mathbf{p}_i - \frac{\mu_i}{M} \sum_{j=0}^N \mathbf{p}_j. \quad (\text{A2})$$

These expressions are valid both classically and quantum mechanically, and the quantum operators satisfy the canonical commutation relations. Note that if one defines internal coordinates with respect to the centre of mass, one has an overcomplete set of internal variables. Therefore, these variables have to satisfy a constraint condition, and do not satisfy the canonical commutation rules, which, however, does not lead to any difficulties [23].

The inverse relations read

$$\begin{aligned} \mathbf{r}_0 &= \mathbf{R} - \sum_{j=1}^N \frac{\mu_j \boldsymbol{\rho}_j}{M} & \mathbf{r}_i &= \mathbf{R} + \boldsymbol{\rho}_i - \sum_{j=1}^N \frac{\mu_j \boldsymbol{\rho}_j}{M} \\ \mathbf{p}_0 &= \frac{\mu_0}{M} \mathbf{P} - \sum_{j=1}^N \boldsymbol{\pi}_j & \mathbf{p}_i &= \frac{\mu_i}{M} \mathbf{P} + \boldsymbol{\pi}_i. \end{aligned} \quad (\text{A3})$$

Finally, the atomic angular momentum can be separated as

$$\mathbf{J}_{\text{atom}} = \sum_{i=0}^N \mathbf{r}_i \times \mathbf{p}_i = \mathbf{R} \times \mathbf{P} + \sum_{i=1}^N \boldsymbol{\rho}_i \times \boldsymbol{\pi}_i \quad \equiv \mathbf{J}_{\text{ext}} + \mathbf{J}_{\text{int}}. \quad (\text{A4})$$

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Optical Pumping of Orbital Angular Momentum of Light in Cold Cesium Atoms

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(Received 25 May 1999)

We present experimental results on the transfer of the orbital angular momentum of light to a system of cold cesium atoms. A nondegenerate four-wave mixing process was used as an indirect tool to observe this transfer. Our experiments show, in particular, that the orbital angular momentum of light can be transferred, via optical pumping in the cold atomic sample, from one beam to another oscillating at a different frequency.

PACS numbers: 32.80.Pj, 42.50.Vk, 42.65.Hw

The Laguerre-Gaussian (LG) laser modes possess well-defined angular momentum that can be decomposed into an orbital component and a spin component associated with its polarization [1]. Optical beams with orbital angular momentum (OAM) include screw topological wave front dislocations or vortices. Vortices appear as spiral phase ramps around a singularity where the phase of the wave is undefined; thus its amplitude must vanish. The order of the singularity multiplied by its sign is referred to as the topological charge of the dislocation.

There has been a great deal of interest in the mechanical and optical effects that light beams with OAM can exert on material systems. This angular momentum has been transferred to macroscopic particles trapped in an optical tweezer causing them to rotate [2,3]. An atom moving in a beam with OAM experiences a torque and an azimuthal shift in its resonant frequency in addition to the usual axial Doppler shift and recoil shift [4]. Also the LG modes are important in laser cooling and trapping experiments. For example, in Ref. [5] such beams were used for high-resolution spectroscopy in a magneto-optical trap, and a novel trapping scheme using LG modes was proposed and experimentally demonstrated in Ref. [6].

The linear propagation of beams with phase front dislocations and also their single-frequency propagation in cubic and photorefractive nonlinear bulk media in single pass and cavity configurations have been studied in [7–10]. The up-conversion by second harmonic generation of light in quadratic nonlinear media has been investigated experimentally in Refs. [11–14]. However, only very limited experimental work has been done with these beams in atomic systems, and has mainly dealt with nonlinear propagation effects [15].

A system of cold atoms as a medium for studying optical processes with beams carrying OAM possesses a number of interesting properties. First, nonlinear effects arising by individual optical transitions in the system of atomic energy levels constitute by itself basic steps that exist in any nonlinear processes in an arbitrary medium.

The OAM of the input beams can be transferred to the atomic system and, hence, modify these nonlinear optical processes. Furthermore, the low temperature of laser cooled atoms, besides allowing to diminish the Doppler broadening of the atomic levels, can also give access to the direct observation of mechanical effects induced by the atom-field interaction [16].

In this Letter, we present experimental results on the nondegenerate four-wave mixing process (NDFWM) performed in cold cesium atoms with optical beams carrying OAM. We use this process as an indirect tool in order to demonstrate that the OAM can be transferred from the optical beam to the system of cold cesium atoms. This effect is similar to the phenomenon of optical pumping induced orientation and alignment in atomic systems due to the spin angular momentum of the photon [17], and to our knowledge corresponds to the first observation of the optical pumping of OAM of light.

We have employed a sample of cold atoms obtained from a vapor cell magneto-optical trap. The trapping beams are provided by a stabilized Ti-sapphire laser, and are red detuned by about 12 MHz from the resonance frequency of the cesium cycling transition $6S_{1/2}$, $F = 4-6P_{3/2}$, $F' = 5$ at $\lambda = 852$ nm, as indicated in Fig. 1(a). The necessary repumping laser is a long external cavity diode laser, which is tuned into resonance with the $6S_{1/2}$, $F = 3-6P_{3/2}$, $F' = 3$ transition. Typically the number of trapped atoms, estimated by measuring the absorption of a weak probe beam by the atomic cloud is of the order of 10^7 atoms. Recently, four-wave mixing experiments in this atomic system were done with beams without OAM [18].

The experimental setup is shown in Fig. 1(b). The forward pump beams F and P with linear polarizations have the same frequency ω_1 , and are incident in the trap forming an angle of $2\theta \approx 2^\circ$. These beams are provided by a grating-stabilized diode laser (LASER1), and are locked in resonance with the noncycling transition $6S_{1/2}$, $F = 4-6P_{3/2}$, $F' = 4$, using an auxiliary saturated

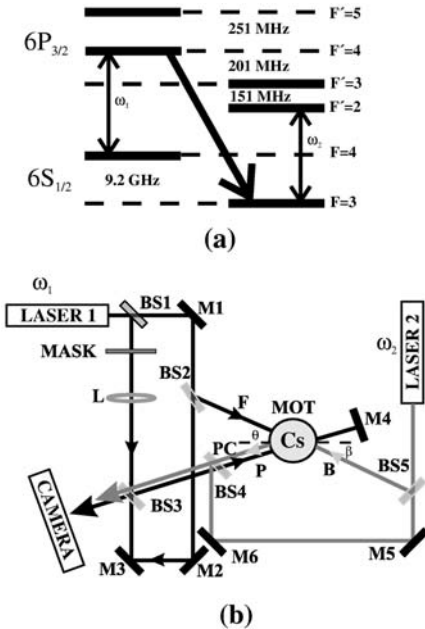


FIG. 1. (a) Hyperfine energy levels of cesium D_2 line. (b) Experimental setup.

absorption signal. Single-charge and double-charge topological screw dislocations were nested in the beam P using two different computer-generated spiral zone plates MASK [19]. A lens L allowed us to separate the beam with a nested vortex from the undiffracted light and high-order Fresnel images.

The beam P is focused so that its waist in the trap is much smaller than the trap size (about 2 mm), while the beam F is collimated to a diameter of the order of 3 mm. To reveal the topological phase structure of the beam P , we used a part of the beam with the frequency ω_1 as a reference beam. The reference beam is reflected from the mirrors $M1$, $M2$, and $M3$, and after a beam splitter BS3 provided an interference pattern with the incident beam P , back reflected from the mirror $M4$. This mirror is removed during the four-wave mixing experiments. The intensity pattern was analyzed by a charge coupled device camera.

Another independent, grating-stabilized diode laser (LASER2) generated the backward pump beam B . The frequency ω_2 of the beam B is set, using other auxiliary reference absorption signal, around the frequency of the cycling transition $6S_{1/2}, F = 3 - 6P_{3/2}, F' = 2$. This beam, also linearly polarized, has a diameter of the order of 3 mm and is incident in the trap satisfying the phase-matching condition, i.e., $\sin\beta = (\omega_1/\omega_2)\sin\theta$. The powers of beams P , F , and B are $P_P = 5 \mu\text{W}$, $P_F = 10 \mu\text{W}$, and $P_B = 200 \mu\text{W}$, respectively. The

generated beam, PC, is therefore nearly counterpropagating to the beam P . Its phase structure was revealed using an interference pattern after a beam splitter BS4 obtained with the reference beam of frequency ω_2 . To measure the phase structure of the incident beam P we stop the beam from the LASER2, while to measure the phase structure of the beam PC we stop the reflection from the mirrors $M2$ and $M4$. The efficiency of the NDFWM process, measured by the ratio P_{PC}/P_P , was of the order of 0.1%.

Figures 2(a) and 2(b) show, respectively, the intensity profile and interferogram of the pump beam P , when the MASK generates into the beam P a single charge phase dislocation. Figures 2(c) and 2(d) show the corresponding distributions for the generated beam PC. As seen, the beam PC includes also a phase front dislocation and hence, an OAM. Owing to the reversed sense of rotation of the interference spiral due to the reflection in mirror $M4$, the topological charge of the beams P and PC are the same. Therefore, as these beams are nearly counterpropagating, they carry opposite OAM.

If the topological charge of the pump beam P is equal to 2, the corresponding intensity profile and interferogram are shown in Figs. 3(a)–3(d). We again observe the generation of the PC beam with the same topological charge as the one of the beam P .

We have modeled our system by considering the incident FWM beams to be of the form $\vec{E}_\mu = \frac{1}{2}\vec{A}_\mu \exp[i(\omega_\mu t - \vec{k}_\mu \cdot \vec{r})] + \text{c.c.}$, with $\mu = F, P, B$; $\omega_F = \omega_P = \omega_1$; $\omega_B = \omega_2$, and $\vec{A}_\mu$ being the corresponding complex field amplitudes. The fields $\vec{E}_F$ and $\vec{E}_B$ are taken as plane waves, so that $\vec{A}_F$ and $\vec{A}_B$ do

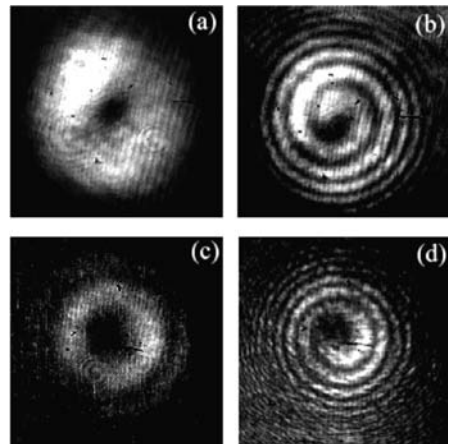


FIG. 2. Observed light distributions and interferograms of the pump beam P [(a) and (b)] and the diffracted beam PC [(c) and (d)]. The topological charge of phase dislocation in the pump beam P is equal to 1 and is the same as that of the beam PC since mirror $M4$ reverses the sense of rotation of the spiral as discussed in the text.

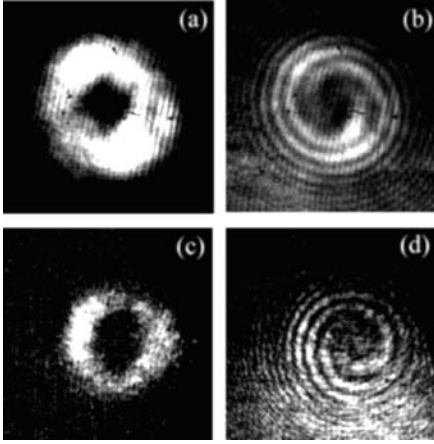


FIG. 3. Same as in Fig. 2 but with the double topological charge of phase dislocation in the pump beam P .

not depend on the coordinates, while for $\vec{E}_P$, the field amplitude of the LG mode at the beam waist, located at $z = 0$ and coincident with the trap position, is given by $\vec{A}_P = \vec{A}_0 \left(\frac{\sqrt{2}r}{w}\right)^l \exp(-il\phi) \exp[-(\frac{r^2}{w^2})] L_p^l(2r^2/w^2)$. Here w is the half-beam width, $\vec{A}_0$ is the beam amplitude, L_p^l is the associated Laguerre polynomial, and r and ϕ are, respectively, the radial and angular coordinates in cylindrical polar coordinate system with its z axis being along the beam propagation direction. The beam P employed in the experiments corresponds to $l = 1$; $p = 0$ (charge 1), and $l = 2$; $p = 0$ (charge 2).

The effect of the trapping and the repumping beams is taken into account through incoherent optical pumping rates specified by γ_T and γ_R , respectively. We have supposed the same relaxation rate γ for all the excited states. We also have assumed that the residual Doppler shift is much smaller than all the relaxation rates, i.e., $\vec{k}_\mu \cdot \vec{v} \ll \gamma, \gamma_R$, where $\vec{v}$ is the atomic velocity. A straightforward calculation using the density matrix in the slowly varying amplitude limit and in the rotating-wave approximation yields for the on-resonance third-order induced coherence at the transition $F = 3 - F' = 2$ the following result [18]:

$$\rho_{32}^{(3)} = \frac{iN\mu_{32}|\mu_{44}|^2}{\hbar^3\gamma^2} \left(\frac{\gamma_T + \gamma}{2\gamma_T + \gamma} \right) \left(\frac{1}{\gamma} + \frac{1}{\gamma_R} \right) A_B A_F A_P^* \times \exp[i(\omega_2 t - (\vec{k}_F + \vec{k}_B - \vec{k}_P) \cdot \vec{r})].$$

In this expression μ_{44} and μ_{32} are the dipole matrix elements of the transitions involved, and N is the total number of atoms in the trap. This induced polarization generates a field oscillating at the frequency ω_2 and propagating along the direction $\vec{k}_{PC} = \vec{k}_F + \vec{k}_B - \vec{k}_P$, under the phase-matching condition. The complex field

amplitude of the generated beam is proportional to the complex conjugated amplitude of the beam P . This can lead to the observation of wave-front rectification and especially in this case to the transfer of the OAM from one beam to another beam with different frequency. Furthermore, the complex amplitude of the generated phase conjugated beam PC has the topological phase defect of the same sign as the one in the beam P , since the induced polarization is proportional to A_P^* , but the propagation directions of the beams P and PC are opposite.

The NDFWM signal can be interpreted in terms of the Bragg diffraction into a transferred population grating. The grating is produced by excitation of the noncycling $6S_{1/2}, F = 4 - 6P_{3/2}, F' = 4$ transition by the beams F and P . The excited-state population grating created by these beams is transferred through spontaneous emission into the $6S_{1/2}, F = 3$ ground-state hyperfine level. The transferred grating is then monitored by the third laser beam B .

In the writing process of the excited-state population grating by absorption of each photon of the beam P , the OAM of this beam is transferred to the atomic system. The angular momentum is stored by the atomic system as the rotational movement of the atoms and is conserved during the spontaneous emission into the $6S_{1/2}, F = 3$ ground-state level. In the reading process by absorption of each photon from the beam B the atomic system generates a photon with OAM in the beam PC. The net OAM transferred to the atomic system depends on the number of photons absorbed in the beam P and the number of photons emitted in the beam PC. Since the atoms in the lower ground state $6S_{1/2}, F = 3$ do not experience any confining force, there is no external torque acting on the atoms to compensate the corresponding change of OAM due to the interaction with the light. Therefore, the atomic sample as a whole should rotate. By taking into account the measured Bragg diffraction efficiency and the absorption of beam P , and the fact that each absorbed and generated photon carries $\pm\hbar$ of OAM, we can roughly estimate the flux of OAM transferred to the atomic system. For the beam P with $l = 1$ this value is of the order of 1×10^{-21} Nm/s. Further investigations to detect directly this atomic motion are currently under way. As we have mentioned before, the transfer of OAM to the cold atomic sample presents a close analogy with the well-known phenomenon of optical pumping induced orientation in atomic systems, a phenomenon which has been used to the production of cold sample of spin-polarized atoms [20].

The present effect could in principle be observed in other nonlinear media, as, for example, in a hot atomic vapor. However, cold atoms with low velocities seem to be essential if one wants to observe the mechanical effects induced on the atomic system due to the OAM exchange with light. Moreover, as light beams carrying

OAM have been suggested to excite vortex states in a Bose-Einstein condensate (BEC) [21], our results, in fact, present a first step towards this goal. Also, as has been demonstrated recently [22], a BEC can be used to perform coherent four-wave mixing of matter waves prepared in well-defined momentum states. The extension of this phenomenon to include the OAM or vortex in the atomic wave packets certainly is of great interest and should provide a better understanding for the coherent matter-wave interaction.

In conclusion, we have observed experimentally the interchange of the OAM between two waves of different frequencies as a result of the optical pumping of the OAM into the system of cold cesium atoms. We hope that this demonstration will trigger further investigation on the coupling between the internal and external degrees of freedom of an atomic system interacting with light, where, in addition to the well-known effect of linear momentum transfer, one also will need to consider its angular counterpart for light carrying OAM.

We are grateful to R. Y. Chiao and J. Rios Leite for stimulating discussions. The work of J. W. R. T. was supported by FINEP, CNPq-Pronex, Brazilian Agencies.

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